BANK OF AMERICA INSTITUTE

BANK OF AMERICA

Economy

Who's still on the move?

23 April 2026

Key takeaways

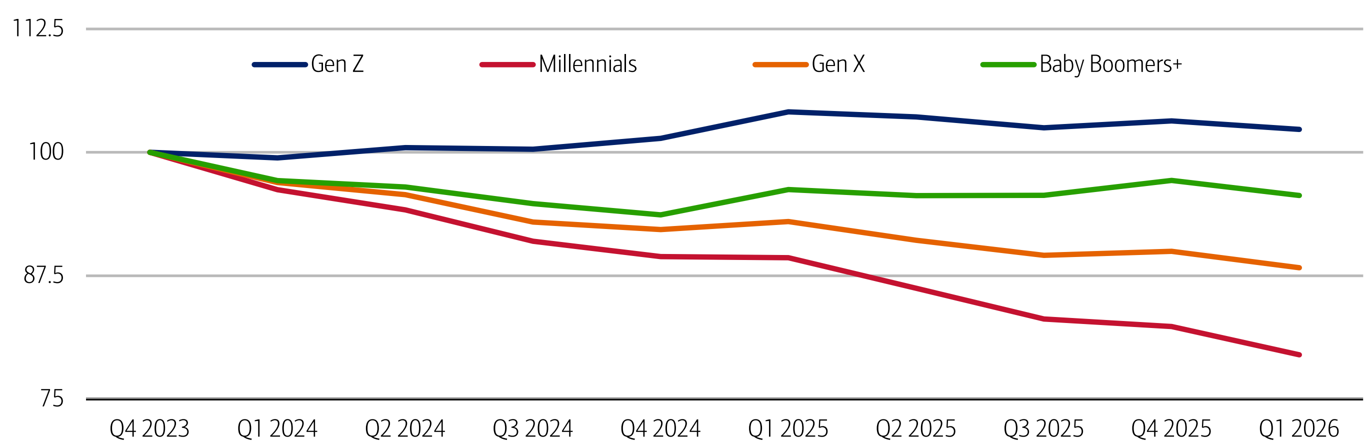
- Overall moving activity has slowed over the past three years, particularly among Millennials and Gen X, while Gen Z remains a notable exception, according to Bank of America account data. Despite a modest pullback over the past year, Gen Z still shows more movers than in 2023, making them the most resilient cohort when it comes to relocating.
- While Gen X and Baby Boomers moved in near lockstep in Q1 2026, often seeking similar metro areas, Gen Z tends to gravitate toward a distinct set of cities including Denver, Minneapolis, Austin and several high-cost West Coast locales, according to Bank of America account data. This suggests that while some Gen Z may be attracted to cities with growing employment and career prospects, lifestyle and culture may also be major drivers.
- Additionally, moving activity has declined far more among lower- and middle-income households than among higher earners over the past three years, with the top 5% by income seeing the smallest drop in movers. This divergence highlights how affordability constraints are increasingly limiting mobility for the less-affluent households.

Moving continues to slow, especially for Millennials and Gen X

Still elevated home prices and interest rates, along with slower hiring, have reduced household mobility within and across cities (read more in On the move: US migration patterns). In fact, the latest Bank of America account data indicates this pattern largely continued through the first quarter of 2026. Millennials accounted for the largest decrease in the number of movers, down nearly 10% year-over-year (YoY), followed by Gen X with an almost 5% decline (Exhibit 1). Baby Boomers have seen a much more muted decline over the past year, although they are still below 2023 levels. Conversely, Gen Z still has more movers compared to 2023, although they have also seen a relatively small decline in the past year.

Exhibit 1: Gen Z has seen an increase in the number of movers, compared to 2023, while all other generations have seen a decrease

The number of people who have moved by generation (rolling four-quarter sum, index full year 2023 = 100)



Source: Bank of America internal data

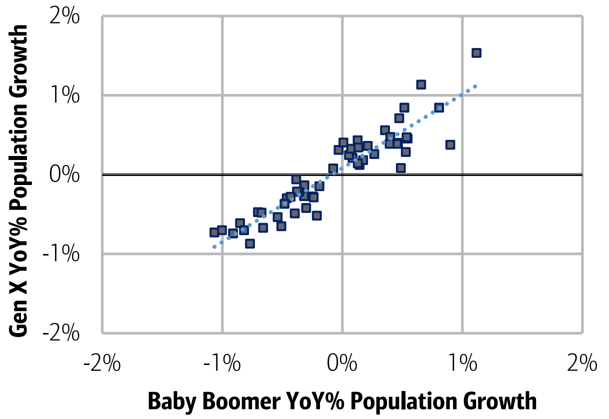
BANK OF AMERICA INSTITUTE

Gen X and Baby Boomers look for some of the same things in a city, unlike Gen Z

Looking across major metropolitan statistical areas (MSAs), Gen X and Baby Boomers move in near lockstep: both largely moving out of and into the same cities (Exhibit 2). Gen Z, on the other hand, has a lot less in common with those generations (Exhibit 3).

Exhibit 2: MSAs that are experiencing Gen X growth and decreases are largely also seeing matching increases and decreases of Baby Boomers

Population growth for Gen X compared to Baby Boomers (Q1 2026, YoY%, each dot represents a major metro area)



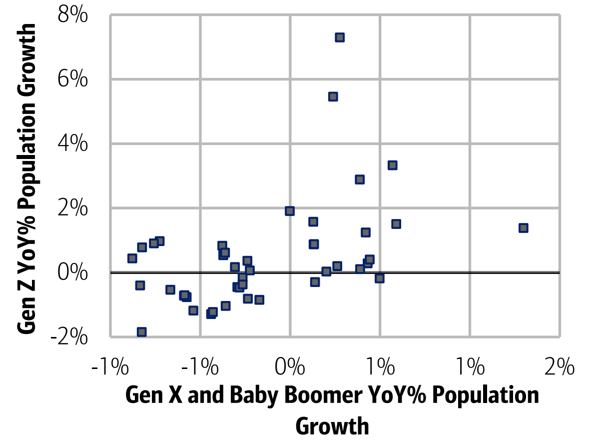
Source: Bank of America internal data

Note: The full list of MSAs is listed in the methodology.

BANK OF AMERICA INSTITUTE

Exhibit 3: Gen Z shows some differentiation in their location choices with Gen X and Baby Boomers

Population growth for Gen Z compared to combined Gen X and Baby Boomer population growth (Q1 2026, YoY%, each dot represents a major metro area)



Source: Bank of America internal data

Note: The full list of MSAs is listed in the methodology.

BANK OF AMERICA INSTITUTE

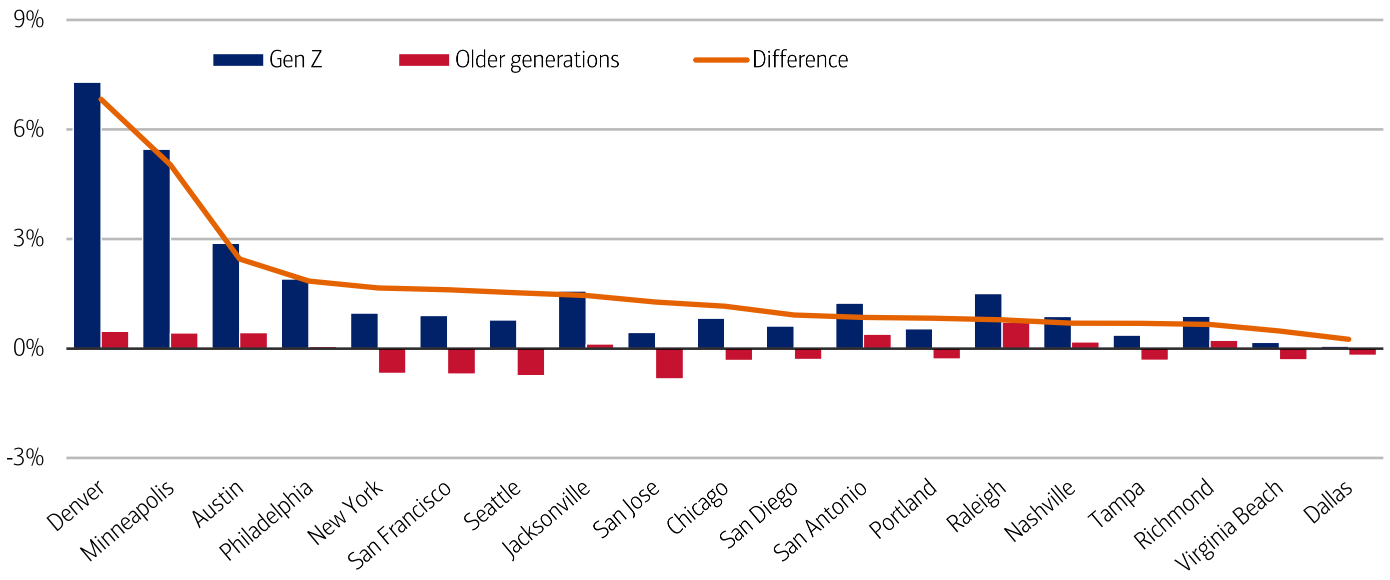
Gen Z is attracted to Denver, Minneapolis and Austin as well as more expensive metros in the West

Examining Bank of America account data through the first quarter of 2026 across metros with the largest divergences helps explain these differences (Exhibit 4). Notably, some cities with the largest relative growth in Gen Z populations compared with older generations – such as Denver and Minneapolis – saw slight YoY declines in the number of jobs created in January 2026, according to data from the Bureau of Labor Statistics (BLS). In our view, this suggests factors besides the labor market (e.g., culture, environment) may attract some younger people to certain cities.

Also noteworthy, higher cost of living may not be as much of a deterrent for Gen Z, according to Bank of America internal data. New York City, San Francisco, Seattle and San Jose all saw increases in their Gen Z population in Q1 2026, while posting decreases among Gen X and Baby Boomers. It's possible that some Gen Z, being earlier in their careers, are more attracted to the opportunities these cities may offer. Meanwhile, older generations may be looking for less expensive locales as they retire, or get closer to retirement, and seek ways to favorably stretch their savings.

Exhibit 4: Gen Z are increasingly moving to Denver, Minneapolis, Austin and Philadelphia relative to older generations

Population growth for Gen Z and older generations (Gen X and Baby Boomers) by major MSA (Q1 2026, YoY%)



Source: Bank of America internal data

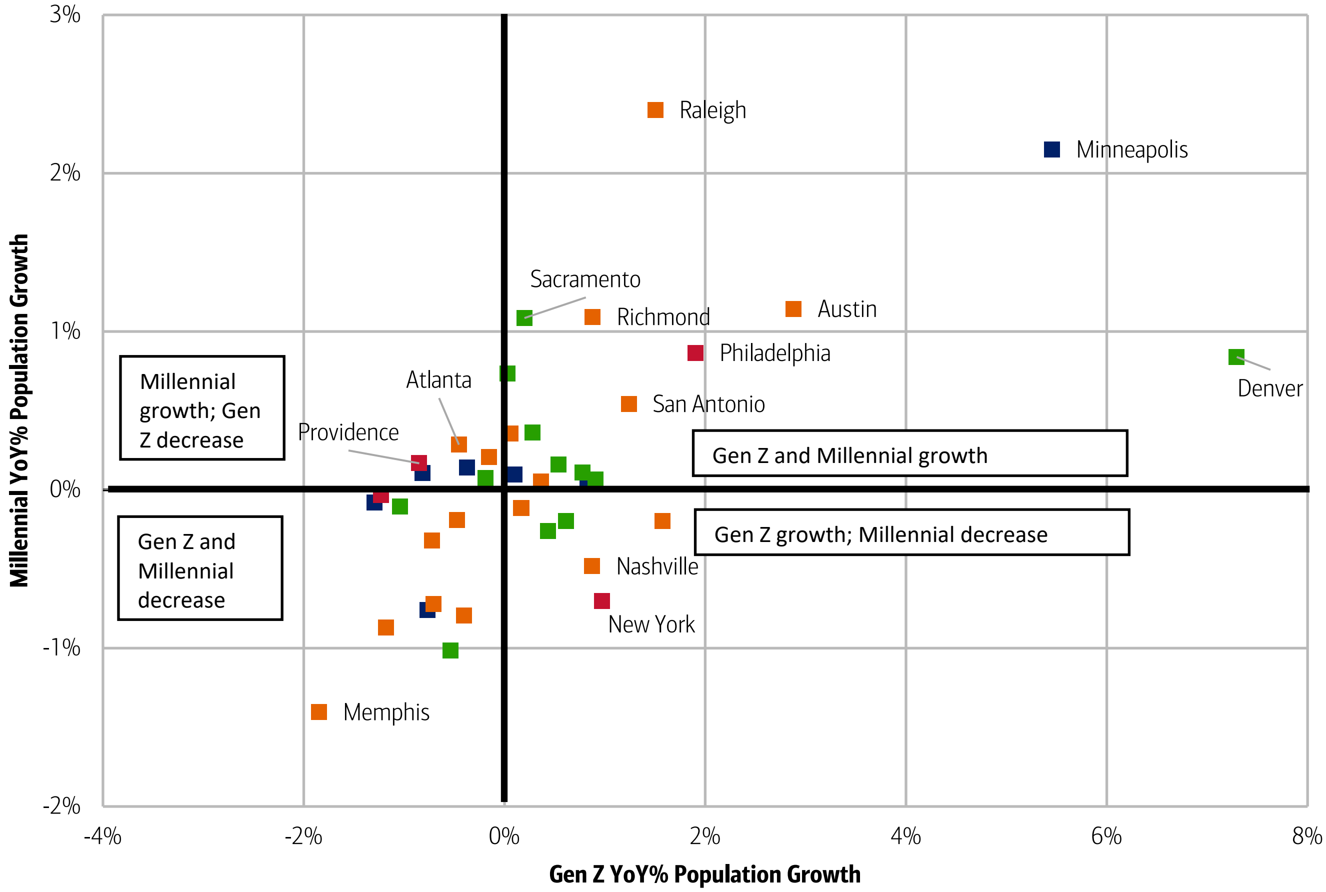
BANK OF AMERICA INSTITUTE

Surprisingly, Gen Z and Millennials weren't leaving or targeting the same cities in the first quarter of 2026, at least not at similar rates (Exhibit 5). However, for those cities that are at the nexus of Gen Z and Millennial population growth, there could be important ramifications. For example, younger people are more likely to be renters (read more in Will younger-gen spending hit a gas-price speed bump?), which may put pressure on the rental markets there.

So, which major metros saw an increase in both generations? In the South: Austin, Raleigh and Richmond stand out, while in the West, tech hubs like Seattle and San Francisco also saw growth, along with a significant bump in Denver. In the Midwest and Northeast, Minneapolis and Philadelphia both had solid increases of Gen Z and Millennial populations.

Exhibit 5: Gen Z and Millennials also differ in location preference with no clear correlation with US Census Region

Population growth for Gen Z compared to Millennial population growth (Q1 2026, YoY%, each dot represents a major metro area) (Red = Northeast, Blue = Midwest, South = Orange, Green = West)



Source: Bank of America internal data

Note: The full list of MSAs is listed in the methodology.

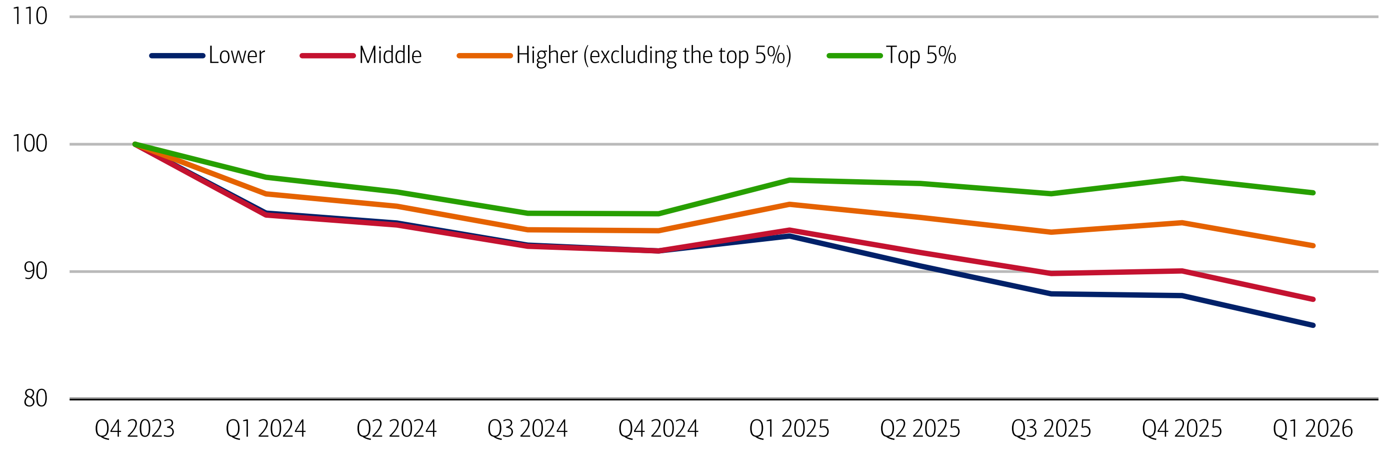
BANK OF AMERICA INSTITUTE

A growing divergence in movers is developing across incomes

Looking across incomes, a gap appears to be developing across income cohorts when it comes to moving. Moving activity fell more sharply over the past three years for lower- and middle-income households than for those with higher incomes, according to Bank of America internal data (Exhibit 6). Notably, higher earners tend to move more than any other group. In fact, the top five percent of households by income had the smallest decline in movers over the past year and three years, likely because they have been less impacted by the rising cost of housing (read more in On the move: Still waiting for the thaw).

Exhibit 6: Lower-income households are moving less than those with higher incomes

The number of people who have moved by household income tercile (rolling four-quarter sum, index 2023 = 100)



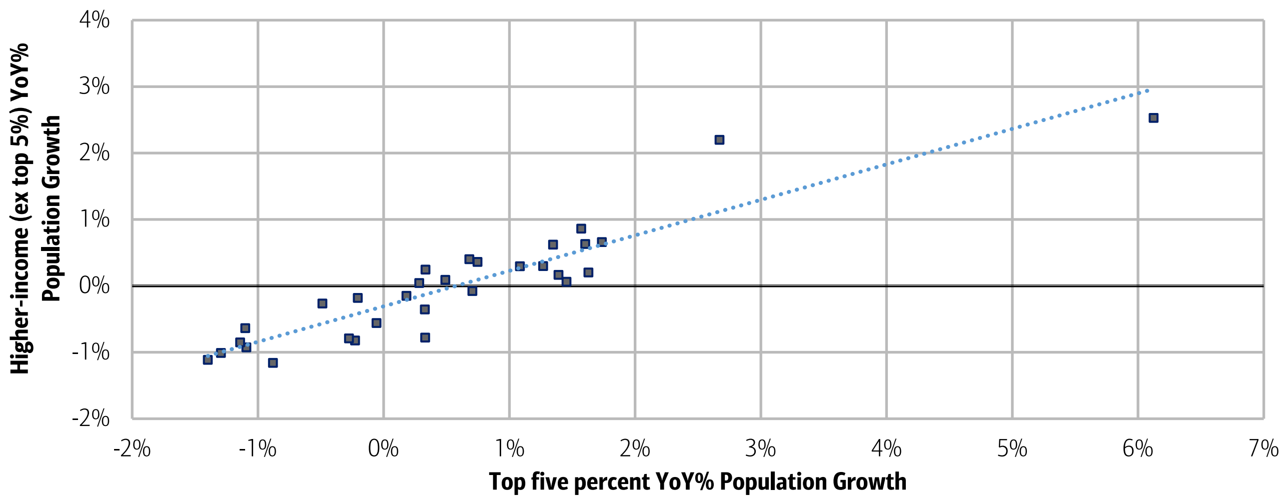
Source: Bank of America internal data

BANK OF AMERICA INSTITUTE

Perhaps unsurprisingly, those in the top five percent by income and the rest of the higher-income group that did move, largely moved into and out of similar cities (Exhibit 7). Both groups left cities including San Francisco, San Jose, Boston, LA and Washington DC while relocating to Philadelphia, Sacramento and Las Vegas. Yet when it comes to overall population growth, Miami, Portland (Oregon) and Houston attracted households in the top five percent of income, while seeing decreases in the rest of the higher-income group.

Exhibit 7: Those with higher incomes tend to favor similar cities

Population growth for higher-income households (excluding the top five percent) compared to the top five percent by income (Q1 2026, YoY%, each dot represents a major metro area)



Source: Bank of America internal data

Note: The full list of MSAs is listed in the methodology.

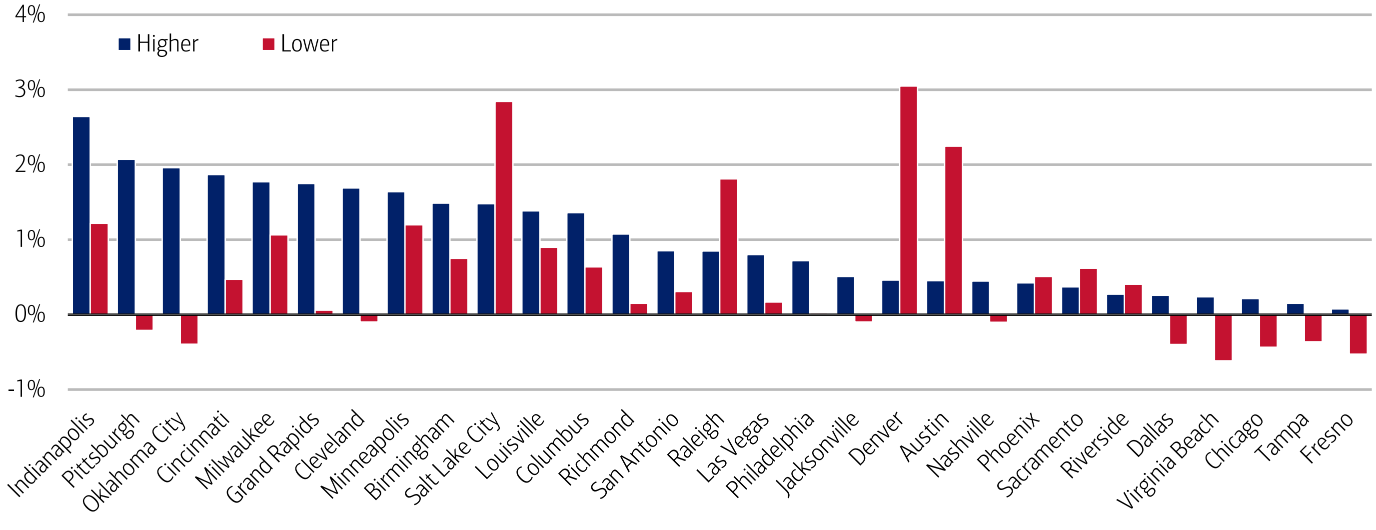
BANK OF AMERICA INSTITUTE

Comparing higher- and lower-income groups, the Midwest stands out because it attracts both (Exhibit 8). For example, Indianapolis, Milwaukee, Minneapolis and Columbus saw solid growth across all incomes. Conversely, Oklahoma City saw stronger population growth of higher-income earners, while actually experiencing a decrease in the number of lower-income households.

Salt Lake City, Denver and Austin saw significantly stronger increases in lower-income households than higher-income ones, possibly reflecting strong population growth for younger generations, many of whom are earlier in their career and therefore earn less money than their older counterparts. Overall, in our view, while the number of movers continues to decline, some cities are seeing growth, even if it is somewhat uneven across income and age groups.

Exhibit 8: The Midwest attracts both lower- and higher-income households

Population growth for higher- and lower-income households by major MSA (Q1 2026, YoY%)



Source: Bank of America internal data

BANK OF AMERICA INSTITUTE

Methodology

Selected Bank of America transaction data is used to inform the macroeconomic views expressed in this report and should be considered in the context of other economic indicators and publicly available information. In certain instances, the data may provide directional and/or predictive value. The data used is not comprehensive; it is based on aggregated and anonymized selections of Bank of America data and may reflect a degree of selection bias and limitations on the data available.

Our analysis for domestic migration pattern is based on the group of Bank of America customers who had an open consumer checking, savings, credit and/or other investment accounts for every quarter between 4Q 2020 and 4Q 2024. Migration pattern is then extracted based on customer home addresses. This methodology yields a fixed sample size of roughly 45 million customers.

Because our data is based on a fixed sample of customers it will not capture the impact of international migration. Instead, our analysis is designed to look at how internal migration in the United States is changing. Accordingly, the overall population movements in the official Census Bureau data, which also accounts for international migration, will not necessarily align with our data in some MSAs, though our data should give similar directional signals.

These changes in address are also used to identify households that have moved in order to capture the spending on moving-related categories for the six-month period before and after a move. To look at this, we use Bank of America internal credit and debit card spending data for households that moved in June over the period 2020-2025. We then determine the average household spending for the 6 months leading up to the move, denoted as "6-" through "1-", the month of the move, denoted as "0," and for the 6 months after the move.

Median mortgage payments for customers who have not moved was also based on this data and include only customers who have not had a change in address.

Any payments data represents aggregated spend from US Retail, Preferred, Small Business and Wealth Management clients with a deposit account or credit card. Aggregated spend include total credit card, debit card, ACH, wires, bill pay, business/peer-to-peer, cash, and checks. This includes rent payments, although wires, cash, and some (mostly paper) checks intended for rent payments may be excluded.

Any **Small Business** payments data represents aggregate spend from Small Business clients with a deposit account or a Small Business credit card. Payroll payments data include channels such as ACH (automated clearing house), bill pay, checks and wire. Bank of America per Small Business client data represents activity spending from active Small Business clients with a deposit account or a Small Business credit card and at least one transaction in each month. Small businesses in this report include business clients within Bank of America and generally defined as under $5mm in annual sales revenue.

Unless otherwise stated, data is not adjusted for seasonality, processing days or portfolio changes, and may be subject to periodic revisions.

The differences between the total and per household card spending growth rate can be explained by the following reasons:

1. Overall total card spending growth is partially boosted by the growth in the number of active cardholders in our sample. This could be due to an increasing customer base or inactive customers using their cards more frequently.
2. Per household card spending growth only looks at households that complete at least five transactions with Bank of America cards in the month. Per household spending growth isolates impacts from a changing sample size, which could be unrelated to underlying economic momentum, and potential spending volatility from less active users.
3. Overall total card spending includes small business card spending while per household card spending does not.
4. Differences due to using processing dates (total card spending) versus transaction date (per household card spending).
5. Other differences including household formations due to young adults moving in and out of their parent's houses during COVID.

Any household consumer deposit data based on Bank of America internal data is derived by anonymizing and aggregating data from Bank of America consumer deposit accounts in the US and analyzing that data at a highly aggregated level. Whenever median household savings and checking balances are quoted, the data is based on a fixed cohort of households that had a consumer deposit account (checking and/or savings account) for all months from January 2019 through the most current month of data shown.

Lower, middle, higher (excluding top 10), and top 10 mortgage payment cuts in Bank of America payments data are based on median monthly mortgage payments in each zip code. These zip codes are then ranked in order from high to low and bucketed according to terciles, with a third of mortgage payments placed in each tercile periodically. The lowest tercile represents "lowest mortgages”, the middle tercile represents “middle mortgages” and the highest tercile “higher mortgages”. The top 10% is then further separated from the highest tercile to denote the top 10% of zip codes by median mortgage payments. The zip codes are reallocated over time, reflecting any number of factors that impact mortgages, including inflation, net domestic migration and shifting supply/demand. The median mortgages payments in each zip code are periodically re-assessed.

Bank of America aggregated credit/debit card spending per household includes spending from active US households only. Only consumer card holders making a minimum of five transactions a month are included in the dataset. Spending from corporate cards are excluded. Data regarding merchants who receive payments are identified and classified by the Merchant Categorization Code (MCC) defined by financial services companies. The data are mapped using proprietary methods from the MCCs to the North American Industry Classification System (NAICS), which is also used by the Census Bureau, in order to classify spending data by subsector. Spending data may also be classified by other proprietary methods not using MCCs.

Where sample sizes allow, the top 50 MSAs by population are included. MSAs outside of the top 50 are aggregated into buckets labeled: Midwest, Northeast, South and West. The top Metropolitan Statistical Areas (MSAs) align to US Census Regions as follows:

- Midwest: Indianapolis, Chicago, Cleveland, Columbus, Detroit, St. Louis, Cincinnati, Kansas City, Milwaukee, Grand Rapids, Louisville, Oklahoma City, Minneapolis
- Northeast: Boston, New York City, Philadelphia, Providence, Pittsburgh, Hartford
- West: Los Angeles, San Francisco, San Jose, San Diego, Seattle, Denver, Las Vegas, Phoenix, Portland, Salt Lake City, Riverside, Fresno, Sacramento
- South: Atlanta, Austin, Baltimore, Charlotte, Dallas, Houston, Jacksonville, Miami, Nashville, Orlando, San Antonio, Tampa, Washington DC, Birmingham, Virginia Beach, Memphis, Raleigh, Richmond

The Sunbelt most commonly refers to the South and Southwestern states of Florida, Georgia, South Carolina, Alabama, Mississippi, Louisiana, Texas, New Mexico, Arizona, Nevada, and California as well as the Southern portion of Colorado, North Carolina, Tennessee, and Utah.

In Exhibit 2, the MSAs that are experiencing both Gen X and Baby Boomer growth are: Atlanta, Baltimore, Boston, Charlotte, Chicago, Dallas, Detroit, Houston, Los Angeles, Miami, New York, Orlando, Portland, San Diego, San Francisco, San Jose, Seattle, St. Louis, Tampa, Washington DC, Kansas City Providence, Virginia Beach, Fresno, Hartford, and Memphis. MSAs that are experiencing decreases in both: Austin, Cleveland, Columbus, Denver, Indianapolis, Las Vegas, Midwest, Nashville, Northeast, Phoenix, San Antonio, South, West, Cincinnati, Milwaukee, Minneapolis, Pittsburgh, Riverside, Sacramento, Birmingham, Grand Rapids, Louisville, Oklahoma City, Raleigh, Richmond, and Salt Lake City. Jacksonville and Philadelphia saw decreasing Baby Boomer populations and growing Gen X populaces.

In Exhibit 3, Atlanta, Baltimore, Boston, Charlotte, Detroit, Houston, Los Angeles, Miami, Orlando, St. Louis, Washington D.C., Kansas City, Providence, Fresno, Hartford and Memphis saw decreasing Gen Z and older generation populations. Austin, Denver, Jacksonville, Midwest, Nashville, Phoenix, San Antonio, South, West, Minneapolis, Riverside, Sacramento, Oklahoma City, Raleigh and Richmond saw increasing Gen Z and older generation populaces. Las Vegas and Northeast saw an increase in older gens and a decrease in Gen Z. Chicago, Dallas, New York, Philadelphia, Portland, San Diego, San Francisco, San Jose, Seattle, Tampa and Virginia Beach saw decreasing older gens and growing Gen Z populations.

In Exhibit 5, the following MSAs saw Gen Z and Millennial growth: Raleigh, Minneapolis, Austin, Richmond, Sacramento, Philadelphia, Denver, Riverside, San Antonio, Phoenix, Dallas, Portland, Seattle, Oklahoma, Chicago, San Francisco and Tampa. The following MSAs saw decreasing populations for both generations: Hartford, St. Louis, Fresno, Houston, Baltimore, Orlando, Boston, Washington D.C., Miami, Los Angeles and Memphis. The following MSAs are growing their Millennial populaces, but have shrinking Gen Z residents: Atlanta, Charlotte, Providence, Kansas City, Detroit, and Las Vegas. The following have growing Gen Z populations and shrinking Millennials: Virginia Beach, San Diego, Jacksonville, San Jose, Nashville and New York.

In Exhibit 7, the following MSAs saw top five percent and other higher income growth: Austin, Chicago, Dallas, Denver, Las Vegas, Midwest, Nashville, Northeast, Orlando, Philadelphia, Phoenix, South, Tampa, West, Riverside and Sacramento. The following MSAs saw a decline in both: Atlanta, Baltimore, Boston, Charlotte, Los Angeles, New York, San Diego, San Francisco, San Jose, Seattle and Washington. The following MSAs saw growth in the top five percent, but declines in the rest of the higher-income cohort: Houston, Miami, Portland (Oregon), and Providence.

Generations, if discussed, are defined as follows:

1. Gen Z, born after 1996;

2. Millennials: born between 1978-1995;

3. Gen Xers: born between 1965-1977;

4. Baby Boomer: 1946-1964

Additional information about the methodology used to aggregate the data is available upon request.

Contributors

Joe Wadford

Economist, Bank of America Institute

Sources

Akshita Jain

Assistant Vice President, Global Risk Analytics

Li Wei

Director, Global Risk Analytics

Disclosures

These materials have been prepared by Bank of America Institute and are provided to you for general information purposes only. To the extent these materials reference Bank of America data, such materials are not intended to be reflective or indicative of, and should not be relied upon as, the results of operations, financial conditions or performance of Bank of America. Bank of America Institute is a think tank dedicated to uncovering powerful insights that move business and society forward. Drawing on data and resources from across the bank and the world, the Institute delivers important, original perspectives on the economy, sustainability and global transformation. Unless otherwise specifically stated, any views or opinions expressed herein are solely those of Bank of America Institute and any individual authors listed, and are not the product of the BofA Global Research department or any other department of Bank of America Corporation or its affiliates and/or subsidiaries (collectively Bank of America). The views in these materials may differ from the views and opinions expressed by the BofA Global Research department or other departments or divisions of Bank of America. Information has been obtained from sources believed to be reliable, but Bank of America does not warrant its completeness or accuracy. These materials do not make any claim regarding the sustainability of any product or service. Any discussion of sustainability is limited as set out herein. Views and estimates constitute our judgment as of the date of these materials and are subject to change without notice. The views expressed herein should not be construed as individual investment advice for any particular person and are not intended as recommendations of particular securities, financial instruments, strategies or banking services for a particular person. This material does not constitute an offer or an invitation by or on behalf of Bank of America to any person to buy or sell any security or financial instrument or engage in any banking service. Nothing in these materials constitutes investment, legal, accounting or tax advice.

Economy

Consumer Checkpoint: The madness of March

10 April 2026

Key takeaways

- Total credit and debit card spending per household rose 4.3% year-over-year (YoY), marking the strongest growth since early 2023, according to Bank of America internal data. Higher gasoline prices powered some of the increase, with spending at the pump surging 16.5% month-over-month (MoM). However, total card spending excluding gasoline still saw healthy growth at 3.6% YoY.
- Higher-income households' spending growth remains well ahead of those in middle- and lower-income groups. While the gap narrowed slightly in March, that's because gas comprises a higher share of lower-income household budgets; discretionary spending growth actually eased in March for lower-income households while rising for other income cohorts.
- Larger tax refunds are so far providing a meaningful short-term boost - especially for discretionary spending and debt paydown - but the increase in refunds skews toward higher-income households and may only temporarily offset rising cost pressures.

Consumer Checkpoint is a regular publication from Bank of America Institute. It aims to provide a holistic and real-time estimate of US consumers' spending and their financial well-being, leveraging the depth and breadth of Bank of America proprietary data. Such data is not intended to be reflective or indicative of, and should not be relied upon as, the results of operations, financial conditions or performance of Bank of America.

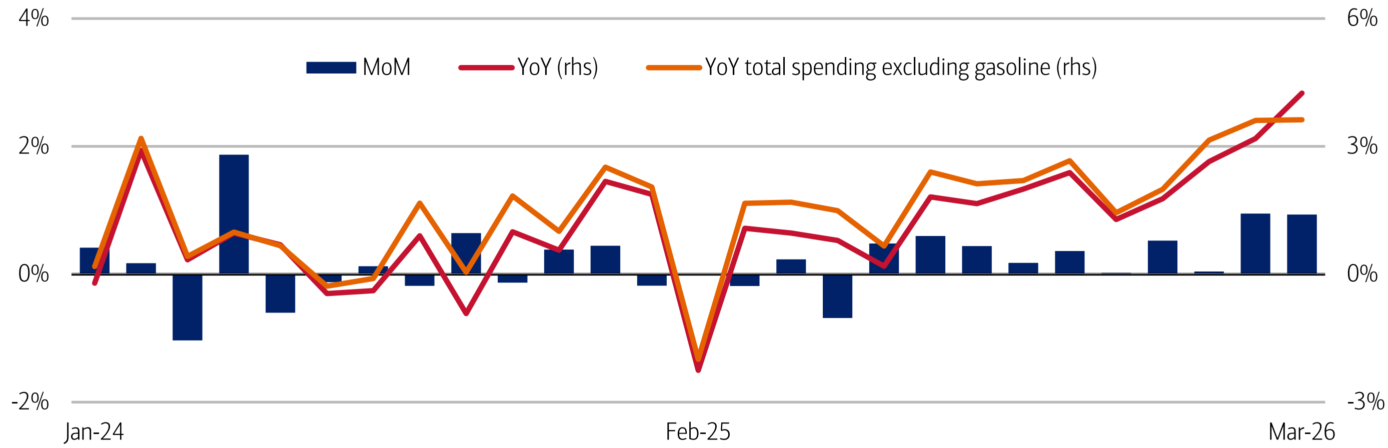
Higher spending, partly driven by gasoline

Total credit and debit card spending per household increased 4.3% year-over-year (YoY) in March 2026, the biggest gain since early 2023. Last month's rise followed the 3.2% YoY jump in February (Exhibit 1). While the increase of more than $1 in gasoline prices in March drove a significant part of the boost in card spending, excluding gasoline, the YoY growth was still a healthy 3.6% in both February and March.

Seasonally-adjusted (SA) spending per household rose 0.9% month-over-month (MoM), after the 1.0% MoM jump in February. Excluding higher gasoline spending, the MoM rise in total spending was a more modest 0.1%.

Exhibit 1: Spending growth strengthened even without the impact of higher gas prices

Total card spending growth per household, based on Bank of America aggregated credit and debit card data (monthly, MoM%, SA) and (monthly, YoY%, non-SA, right-hand side (rhs))



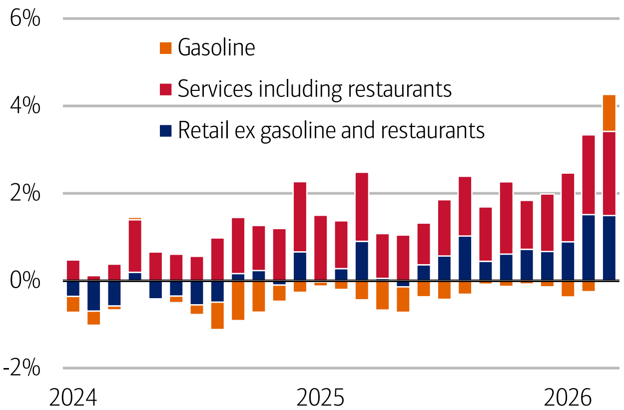
Source: Bank of America internal data

BANK OF AMERICA INSTITUTE

Looking at the composition of YoY card spending growth in March, rising gasoline spending accounted for only 0.8 percentage points (pp), while retail (excluding gas and restaurants) contributed nearly double that amount (Exhibit 2). Services spending made the strongest contribution to growth, at 1.9 pp. Diving deeper within overall services, discretionary spending has been solid since the beginning of the year, with restaurants, travel, tourism and leisure all making positive contributions to YoY spending growth (Exhibit 3).

Exhibit 2: Increases in retail and services spending accounted for the majority of YoY growth

Contribution to YoY total credit and debit card spending growth by category, based on Bank of America card data (monthly, SA, percentage points contribution)

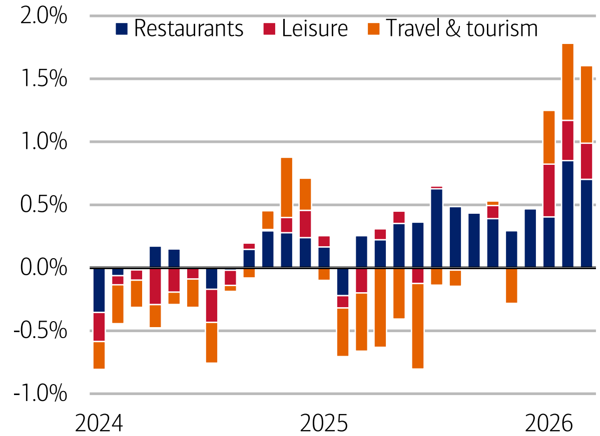


Source: Bank of America internal data

BANK OF AMERICA INSTITUTE

Exhibit 3: Discretionary services categories continued to contribute to YoY spending growth

Contribution to YoY services growth by discretionary services category (monthly, SA, percentage points contribution)



Source: Bank of America internal data

Note: Discretionary services include restaurants, leisure, and travel & tourism.

BANK OF AMERICA INSTITUTE

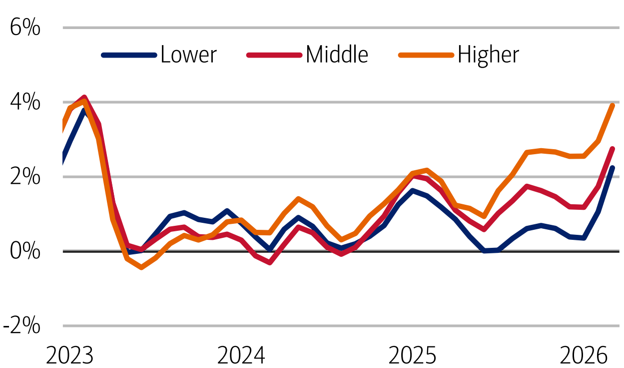
Lower-income households trim discretionary spending as gas takes bigger bite

Looking across income cohorts, lower-income households' total card spending rose to 2.2% YoY in March, narrowing the gap with other income groups for the second month in a row. By contrast, higher-income households' spending growth was much stronger, up 3.9% YoY, while those in middle-income group gained 2.8% YoY.

However, the rise in lower-income households' YoY card spending is largely because gasoline makes up a larger share of spending for those with smaller incomes. Consequently, those with lower incomes curbed discretionary purchases (spending outside of groceries, utilities and gasoline) last month, and their YoY spending growth on discretionary goods dropped back relative to increases seen among middle- and higher-income households. (Exhibit 5). Although it's possible that some of this easing is due to the fading effects of tax refund-driven spending.

Exhibit 4: Higher-income households' spending rose to 3.9% YoY in March compared to 2.2% YoY for lower-income households

Total credit and debit card spending per household, according to Bank of America card data, by household income terciles (3-month moving average, YoY%, SA)

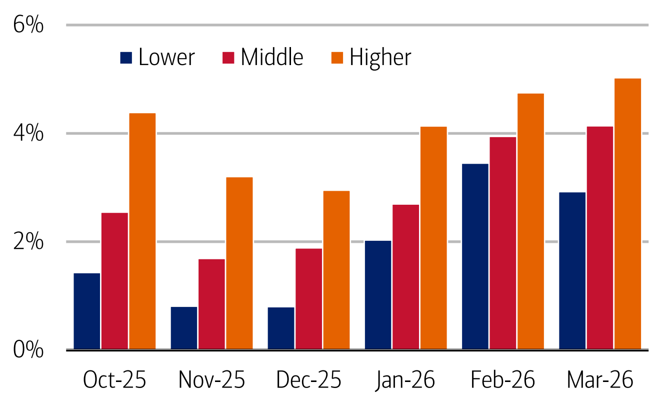


Source: Bank of America internal data

BANK OF AMERICA INSTITUTE

Exhibit 5: YoY discretionary spending growth was slightly more muted for lower-income households compared to February

Discretionary card spending per household, according to Bank of America card data, by household income terciles (YoY%, SA)



Source: Bank of America internal data

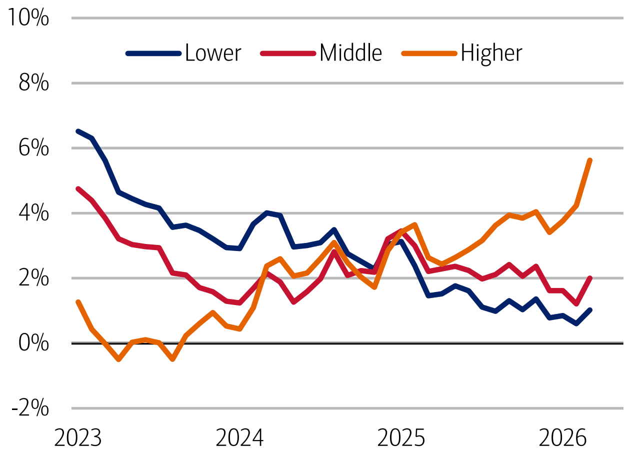
BANK OF AMERICA INSTITUTE

Beyond pain at the gas pump, another reason lower- and middle-income households may be tightening their belts: their earnings growth continues to lag behind those with higher incomes.

Higher-income households' after-tax wage and salary growth rose sharply in March, to 5.6% YoY – the fastest YoY growth since August 2021 – up from 4.2% in February, according to Bank of America deposit data. And while there was a rebound in middle- and lower-income households' after-tax wage growth in March, it was smaller, to 2.0% and 1.0% YoY, respectively. As a result, the divergence with higher-income households widened to the largest gap we've seen since our data series began in 2015 (Exhibit 6).

Exhibit 6: In March, higher-income households' after-tax wage growth rose to 5.6% YoY, while lower-income households' wage growth was 1.0% YoY

After-tax wage and salary growth by household income terciles, based on Bank of America aggregated consumer deposit account data (3-month moving average, YoY%, SA)

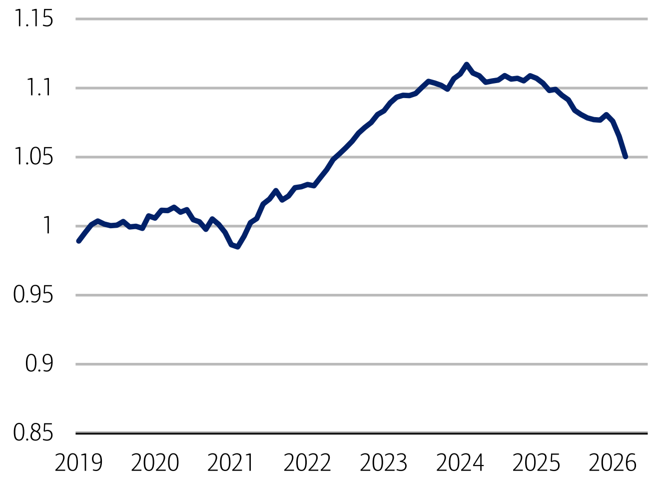


Source: Bank of America internal data

BANK OF AMERICA INSTITUTE

Exhibit 7: The current "K" shape in wage growth has unwound some of the previous relative gains of lower-income households

Lower-income households' after-tax wages and salaries relative to those of higher-income households (yearly, index 2019=1)



Source: Bank of America internal data

BANK OF AMERICA INSTITUTE

As noted in our March Employment Report, the gains among higher-income households are likely, in part, the result of bigger bonuses (read more in The Institute Employment Report: March 2026). By contrast, bonuses declined for lower- and middle-income households. While the impact of this may fade as we move through the year, the persistence in the "K" shape between higher-income wage growth and other cohorts suggests underlying wage dynamics may continue to favor those who earn more. As a result, the "K" shape in spending is also likely to persist.

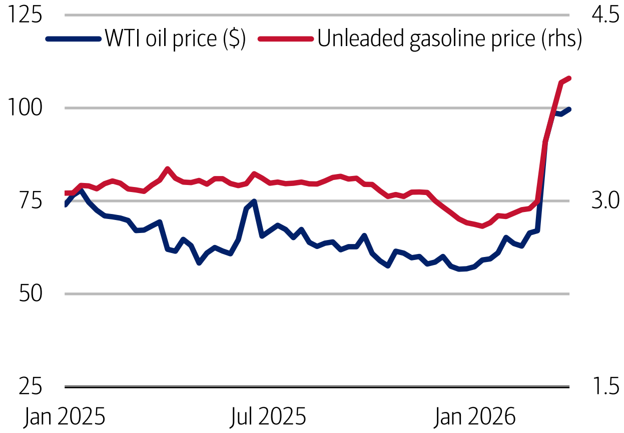
One piece of good news: the wage gap remains smaller than it was before the pandemic. That said, faster pay gains for higher earners are chipping away at the progress lower-income households made from 2020 to 2024 (Exhibit 7).

Will gasoline put the brakes on overall discretionary spending?

The rapid rise in gasoline prices in March – from just below $3 a gallon nationally to around $4 by the end of the month (Exhibit 8) was reflected in a spending increase at the pump. According to Bank of America card data, gasoline spending per household surged 16.5% MoM in March. Interestingly, we also saw a *rise* in gasoline transactions despite higher prices - in our view likely as drivers tried to get ahead of price rises by filling up early (Exhibit 9).

Exhibit 8: Both oil and gasoline prices have surged in the past month…

West Texas Intermediate (WTI) oil price (weekly, $ per barrel) and national average regular unleaded gasoline prices (weekly, $ per gallon, rhs)

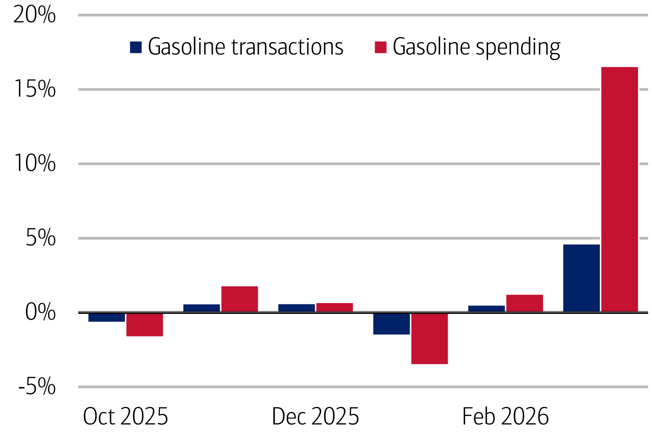


Source: Bloomberg

BANK OF AMERICA INSTITUTE

Exhibit 9: …while both gasoline transactions and spending also rose in March

Gasoline transactions and spending per household, according to Bank of America card data (monthly, % MoM, SA)



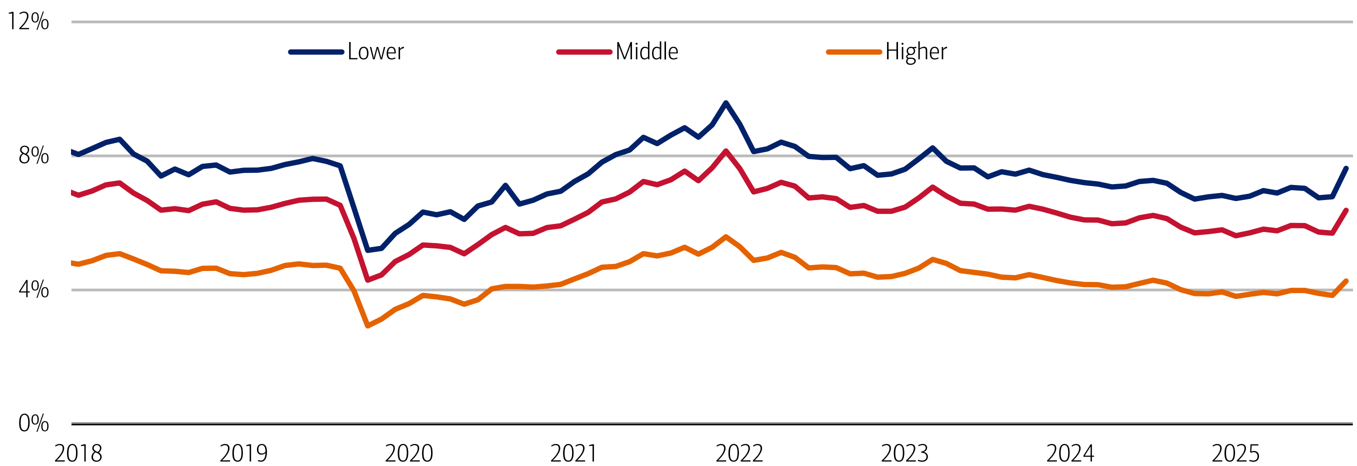
Source: Bank of America internal data

BANK OF AMERICA INSTITUTE

While the duration of the conflict is highly uncertain – as is the impact on oil prices – if higher gas prices are sustained for a prolonged period all households will be impacted, but lower-income families are most exposed. Household monthly gasoline spending for this cohort was equivalent to around 8-10% of their total card spending throughout most of the past eight years, roughly twice that of those with higher incomes (Exhibit 10).

Exhibit 10: For lower-income households, gasoline spending comprised around 8% of their total card spending in March

Gasoline spending relative to total card spending per household, by household income terciles (monthly, %)



Source: Bank of America internal data

BANK OF AMERICA INSTITUTE

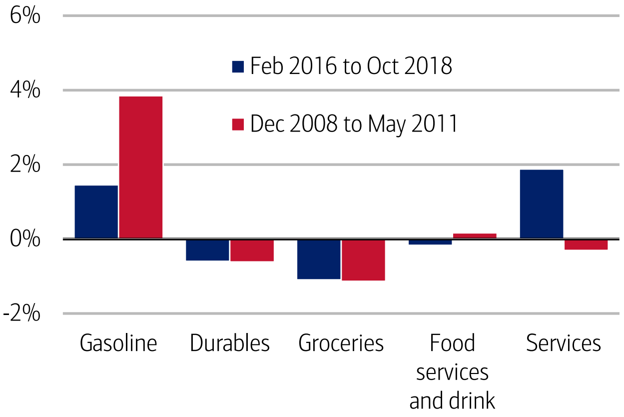
What areas of discretionary spending might households pull back on if higher gasoline prices remain for some time?

Focusing on two periods where the gasoline spending share rose significantly – 2009 to 2011 and 2016 to 2018 – we find that both durable goods and grocery spending shares saw downward pressure (Exhibit 11). In our view, the latter is consistent with consumers trading down in their grocery spending when they face cost pressures in other areas of their necessity outlays.

Looking across the full 2005 to 2026 period, excluding the volatile pandemic years (2020-2022), we also see some tendency for the share of total card spending on services to have been negatively correlated with the share of gasoline in total card spending (Exhibit 12). Within this, it appears spending on food services and drink also has some modest negative relationship.

Exhibit 11: In two previous periods in which gasoline spending increased, spending on durables and groceries were negatively impacted

Changes in the share of select categories relative to total card spending (February 2016-October 2018 and December 2008-May 2011, pp)

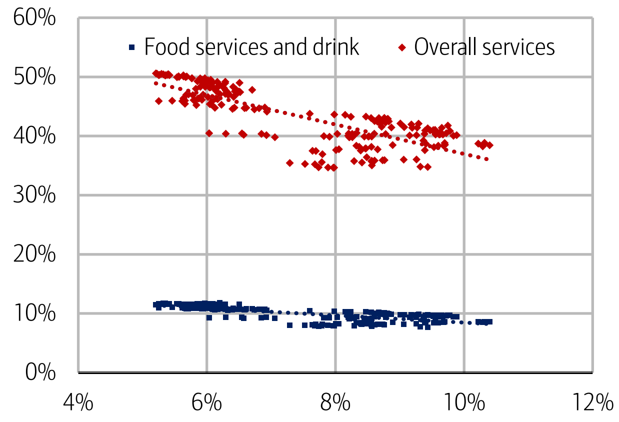


Source: Bank of America internal data

BANK OF AMERICA INSTITUTE

Exhibit 12: There appears to be some negative relationship between services spending and gasoline

Share of gasoline spending in total card spending (x-axis) versus shares of overall services spending (y-axis, red markers) and food services and drink spending (y-axis, blue markers) (monthly, 2005-2025)



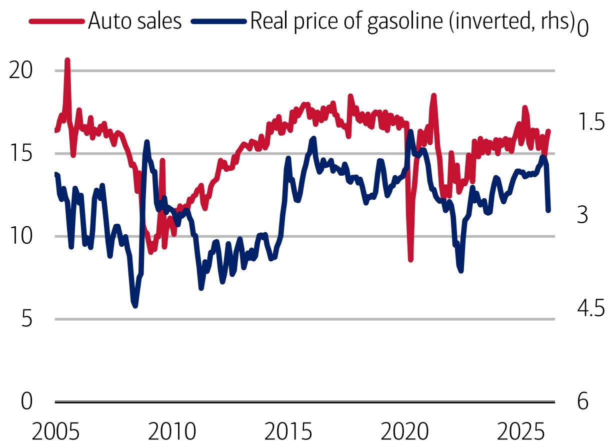
Source: Bank of America internal data

BANK OF AMERICA INSTITUTE

Another area of discretionary spending that would be vulnerable to higher gasoline prices is autos, but consumers don't appear to be pulling back just yet. In March, sales were solid. Furthermore, the current price of gasoline after adjusting for inflation is not as high as in the post-global financial crisis or post-pandemic periods and, more broadly, auto sales have been relatively flat over the last two years (Exhibit 13).

Exhibit 13: The real price of gasoline is below previous peaks

Real gasoline price (rhs, inverted, 2015 prices) and auto sales (seasonally-adjusted annualized rate (SAAR), millions)

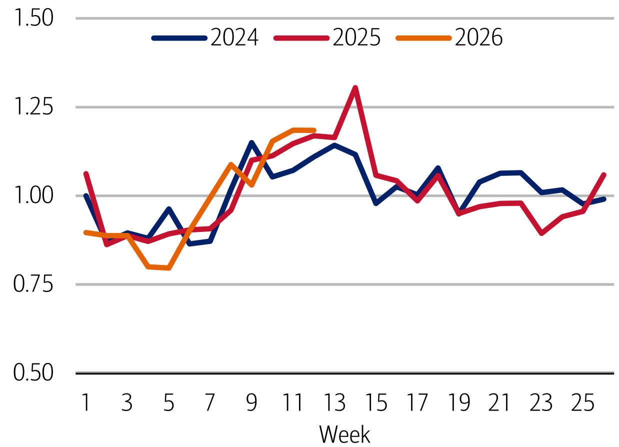


Source: Bloomberg

BANK OF AMERICA INSTITUTE

Exhibit 14: Large auto payments did not weaken over March

Large* auto payments, 2024-2026 (weekly, index January 2024=1)



Source: Bank of America internal data

Note: Large payments are those over $2,000 and where no payment greater than $600 was observed in the previous two months. Data is across payment channels (credit and debit, automated clearing house (ACH) and wire).

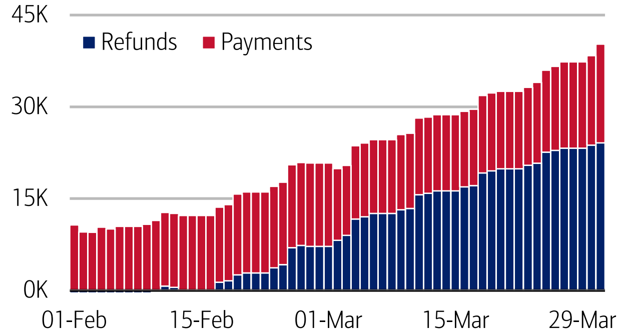
BANK OF AMERICA INSTITUTE

Weekly data on the number of Bank of America customers making large payments (above $2,000) to auto firms and vehicle finance providers – which we are using as a proxy for car sales – also shows no sign of a weakening over the course of March (Exhibit 14).

Higher tax refunds may help counter pain at the pump

One relief for households navigating higher gasoline prices is coming from higher tax refunds and lower tax payments (Exhibit 15). BofA Global Research notes that the lower payments – resulting from the One Big Beautiful Bill Act (OBBBA) – may benefit higher-income households more than anyone else. And Bank of America deposit data confirms that these households are also seeing a bigger increase in tax refunds as well (Exhibit 16).

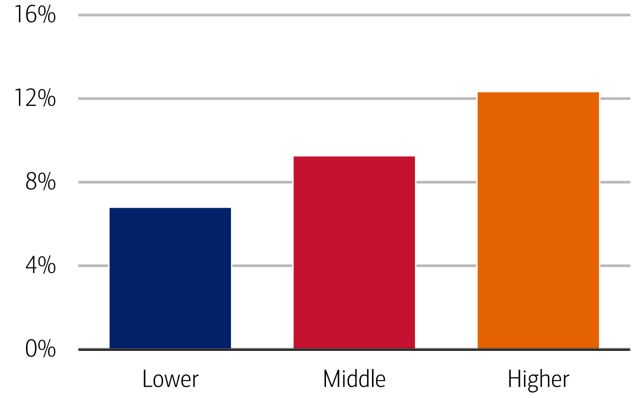
Exhibit 15: Consumers are benefitting from higher refunds and lower tax payments
Federal tax refunds and individual tax payments (daily, year-over-year difference)



Source: US Treasury Department
Note: Tax refunds are the difference between 2026 and 2025 levels. Tax payment differences below zero are considered a positive, as they indicate less tax payments than last year.

BANK OF AMERICA INSTITUTE

Exhibit 16: Average refunds continued to be up for all income cohorts, though most for higher-income households
Average tax refund by household income tercile through April 4 compared to similar period last year (aggregate, YoY%)



Source: Bank of America internal data

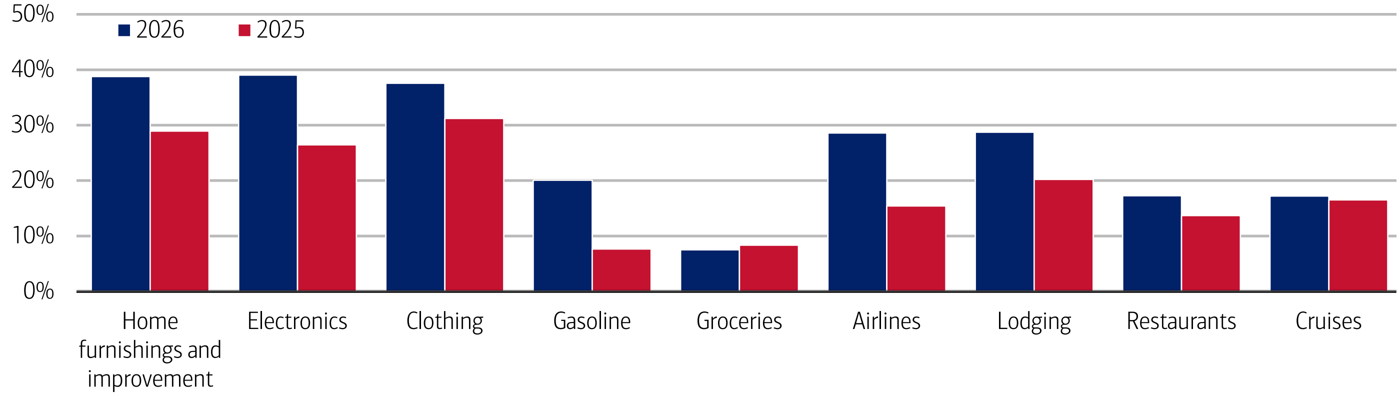
BANK OF AMERICA INSTITUTE

Early tax filers are spending on home improvement, electronics and clothing

Comparing Bank of America payments data on spending before and after receipt of a tax refund shows early refund recipients have so far tended to favor discretionary retail categories like home improvement, electronics and clothing over necessities (groceries) or discretionary travel (airlines, lodging) (Exhibit 17).

Compared to the boost from refunds in 2025, most discretionary categories are seeing a larger tax refund-fueled bump so far this year, which is consistent with the average refund being higher. And while spending at gasoline stores saw a larger boost this year, in our view it more likely reflects the timing of higher gasoline prices as opposed to people using their tax refunds to stock up on gas.

Exhibit 17: Earlier filers are increasing their spending across many discretionary retail and service categories
Spending across all payment channels in the three weeks after receipt of tax refund compared to average of three weeks before, by select retail and service categories (weekly, % change)



Source: Bank of America internal data
Note: 2026 spending trends include tax refunds up to March 6, 2026, compared to the entire tax period for 2025.

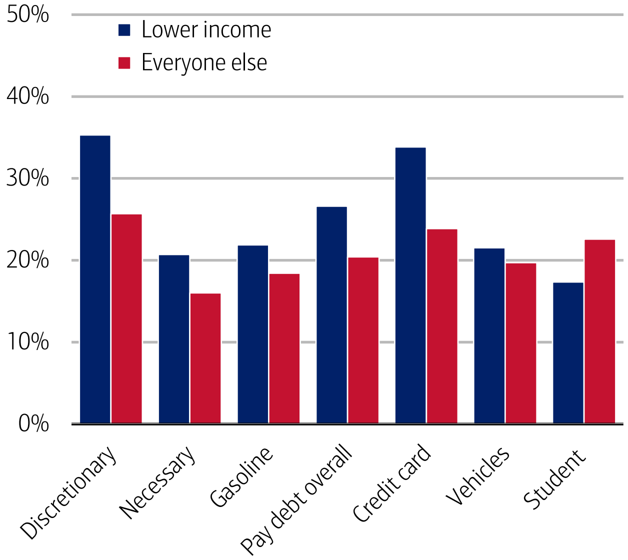
BANK OF AMERICA INSTITUTE

While all income groups have spent more on necessities thanks to higher refunds in 2026, lower-income households experienced a relatively larger boost in discretionary spending than other groups (Exhibit 18). However, in our view, it's possible that some of these boosts have been moderated slightly due to higher gasoline prices.

A similar pattern is evident in debt payments. Lower-income households, in particular, are using their tax refunds to cut debt by increasing their credit card payments. Thus, this tax season has allowed some of them to meaningfully repair their household balance sheets. Additionally, our data shows another potential use for tax refund increases: the extra cash could help buffer the *increases* in gasoline spending for at least five months (Exhibit 19).

Exhibit 18: Lower-income households have seen larger refund-driven discretionary spending and debt payment boosts relative to middle- and higher-income households

Spending across all payment channels in the three weeks after receipt of tax refund compared to average of three weeks before for select categories by customer income tercile (2026, % change)



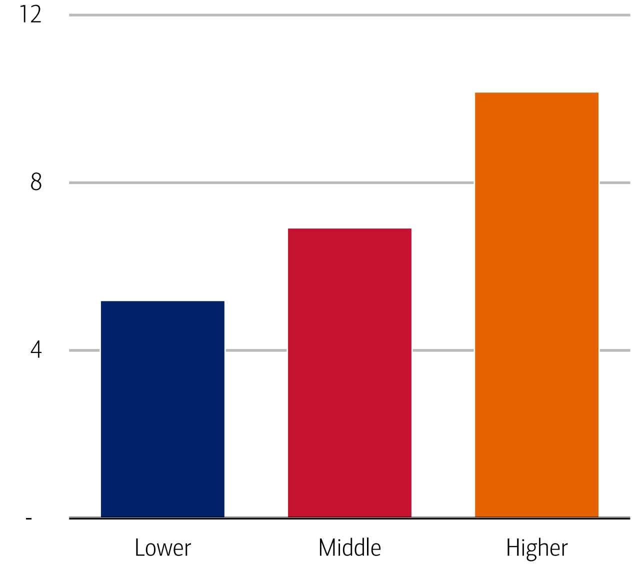
Source: Bank of America internal data

Note: Includes tax refunds up to March 6, 2026. Debt payments include internal (BAC) and external debt payments.

BANK OF AMERICA INSTITUTE

Exhibit 19: Tax refund increases could offset at least five months of increased gas prices

Time that average increase in refunds could offset average increase in gasoline spending by household income tercile (2026, months)



Source: Bank of America internal data

Note: Gas spending increases are based on the per household increase seen in March 2026 compared to March 2025 for households who have gas spending. Average increase in refunds are based on the differences calculated as of April 4, 2026 compared to a similar period in 2025.

BANK OF AMERICA INSTITUTE

Methodology

Selected Bank of America transaction data is used to inform the macroeconomic views expressed in this report and should be considered in the context of other economic indicators and publicly available information. In certain instances, the data may provide directional and/or predictive value. The data used is not comprehensive; it is based on **aggregated and anonymized** selections of Bank of America data and may reflect a degree of selection bias and limitations on the data available.

Any payments data represents aggregated spend from US Retail, Preferred, Small Business and Wealth Management clients with a deposit account or credit card. Aggregated spend include total credit card, debit card, ACH, wires, bill pay, business/peer-to-peer, cash, and checks.

Any **Small Business** payments data represents aggregate spend from Small Business clients with a deposit account or a Small Business credit card. Payroll payments data include channels such as ACH (automated clearing house), bill pay, checks and wire. Bank of America per Small Business client data represents activity spending from active Small Business clients with a deposit account or a Small Business credit card and at least one transaction in each month. Small businesses in this report include business clients within Bank of America and generally defined as under $5mm in annual sales revenue.

Unless otherwise stated, data is not adjusted for seasonality, processing days or portfolio changes, and may be subject to periodic revisions.

The differences between the total and per household card spending growth rate (if discussed) can be explained by the following reasons:

1. Overall total card spending growth is partially boosted by the growth in the number of active cardholders in our sample. This could be due to an increasing customer base or inactive customers using their cards more frequently.

2. Per household card spending growth only looks at households that complete at least five transactions with Bank of America cards in the month. Per household spending growth isolates impacts from a changing sample size, which could be unrelated to underlying economic momentum, and potential spending volatility from less active users.

3. Overall total card spending includes small business card spending while per household card spending does not.

4. Differences due to using processing dates (total card spending) versus transaction date (per household card spending).

5. Other differences including household formations due to young adults moving in and out of their parent's houses during COVID.

Any household consumer deposit data based on Bank of America internal data is derived by anonymizing and aggregating data from Bank of America consumer deposit accounts in the US and analyzing that data at a highly aggregated level. Whenever median household savings and checking balances are quoted, the data is based on a fixed cohort of households that had a consumer deposit account (checking and/or savings account) for all months from January 2019 through the most current month of data shown.

Bank of America aggregated credit/debit card spending per household includes spending from active US households only. Only consumer card holders making a minimum of five transactions a month are included in the dataset. Spending from corporate cards are excluded. Data regarding merchants who receive payments are identified and classified by the Merchant Categorization Code (MCC) defined by financial services companies. The data are mapped using proprietary methods from the MCCs to the North American Industry Classification System (NAICS), which is also used by the Census Bureau, in order to classify spending data by subsector. Spending data may also be classified by other proprietary methods not using MCCs.

We consider a measure of services necessity spending that includes but is not limited to childcare, rent, insurance, insurance, public transportation, and tax payments. Discretionary services includes but is not limited to charitable donations, leisure travel, entertainment, and professional/consumer services. Discretionary retail includes but is not limited to general merchandise, miscellaneous, clothing, electronics, furniture. It excludes categories like groceries and gasoline. Holiday spending is defined as items in which spending in the November-December period is usually at least 20% of total annual spending on the category. Value and premium grocers are determined judgmentally on a proprietary analysis of merchants by equity analysts.

Durables spending is defined as spending on electronics, building materials, auto and furniture. Premium durables spending is based on a selection of retailers who are judged to sell relatively higher value products. Conversely, value durables spending is based on a selection of retailers who are judged to sell relatively lower value products.

For analysis looking at higher value transactions (including durables), we consider a value per transaction threshold estimated with reference to the top 30% of transactions by value in 2024. The share of higher value transactions is then the number of transactions above this threshold as a percentage of total transactions over time.

Lower, middle and higher household income cuts in Bank of America credit and debit card spending per household, and consumer deposit account data are based on quantitative estimates of each households' income. These quantitative estimates are bucketed according to terciles, with a third of households placed in each tercile periodically. The lowest tercile represents 'lower income', the middle tercile represents 'middle income' and the highest tercile 'higher income'. The income thresholds between these terciles will move over time, reflecting any number of factors that impact income, including general wage inflation, changes in social security payments and individual households' income. The income and tercile in which a household is categorised are periodically re-assessed.

Major grocery categories include sugar and sweets, juices and other non-alcoholic beverages, bakery products, processed fruits and vegetables, fresh fruit and vegetables, coffee and tea, fats and oils, milk, cereal and cereal products, other, cheese, and meats, poultry and fish, Other includes soups, snacks, frozen and freeze-dried prepared foods, and spices, seasonings, and condiments.

Generations, if discussed, are defined as follows:

1. Gen Z, born after 1995
2. Younger Millennials: born between 1989-1995
3. Older Millennials: born between 1978-1988
4. Gen Xers: born between 1965-1977
5. Baby Boomer: 1946-1964
6. Traditionalists: pre-1946

Any reference to card spending per household on gasoline includes all purchases at gasoline stations and might include purchases of non-gas items.

Additional information about the methodology used to aggregate the data is available upon request.

Contributors

David Michael Tinsley

Senior Economist, Bank of America Institute

Joe Wadford

Economist, Bank of America Institute

Liz Everett Krisberg

Head of Bank of America Institute

Taylor Bowley

Economist, Bank of America Institute

Vanessa Cook

Content Strategist, Bank of America Institute

Lynelle Huskey

Analyst, Bank of America Institute

Sources

Li Wei

Director, Global Risk Analytics

Yan Peng

Vice President, Global Risk Analytics

Akshita Jain

Assistant Vice President, Global Risk Analytics

Jon Kaplan

Senior Vice President, Consumer & Small Business Products and Analytics

Stephen Stancil

Senior Vice President, Consumer & Small Business Products and Analytics

Disclosures

BANK OF AMERICA INSTITUTE

BANK OF AMERICA

Economy

The Institute Employment Report: March 2026

08 April 2026

Key takeaways

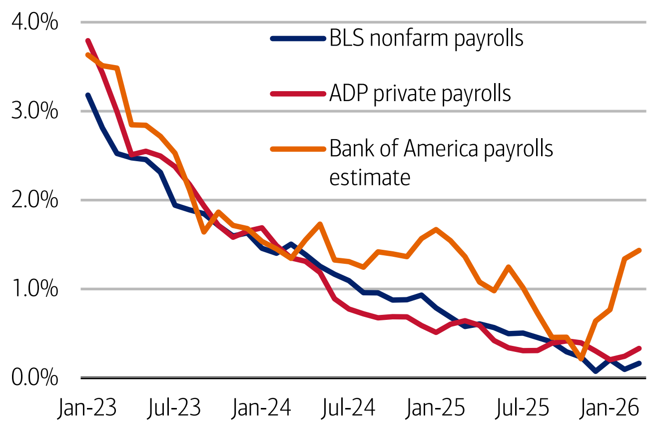
- An estimate of payrolls growth based on Bank of America customer deposit account data continues to suggest a reacceleration in payrolls growth, with year-over-year (YoY) gains rising to 1.4% in March - back in line with early-2025 momentum.
- But while overall jobs growth has improved, after-tax wage growth is increasingly "K-shaped," with higher-income households seeing wage growth of 5.6% YoY in March, versus just 1.0% and 2.0% for lower- and middle-income groups, respectively, according to Bank of America customer account data. This is the widest gap in growth rates since 2015.
- In our view, a surge in bonuses among higher-income households is likely amplifying wage dispersion early in 2026, with solid YoY growth in bonuses for higher-income households while YoY growth in bonuses for middle- and lower-income cohorts remain negative. But even if this effect fades over this year, underlying wage dynamics continue to favor the top end.

Payrolls growth showed continued momentum in March…

Bank of America deposit account data suggests the significant recovery in payrolls growth seen in February continued into March, while unemployment payments growth has eased. However, the "K" shape in wage growth grew wider.

We use Bank of America consumer deposit data to estimate a payrolls series by looking at how the number of customer accounts receiving a paycheck is changing (see Methodology). This data can be fairly noisy, partly due to seasonal variation. However, looking at the three-month moving average, Exhibit 1 suggests that the year-over-year (YoY) growth in our measure rose to 1.4% in March 2026, up from 1.3% YoY in February.

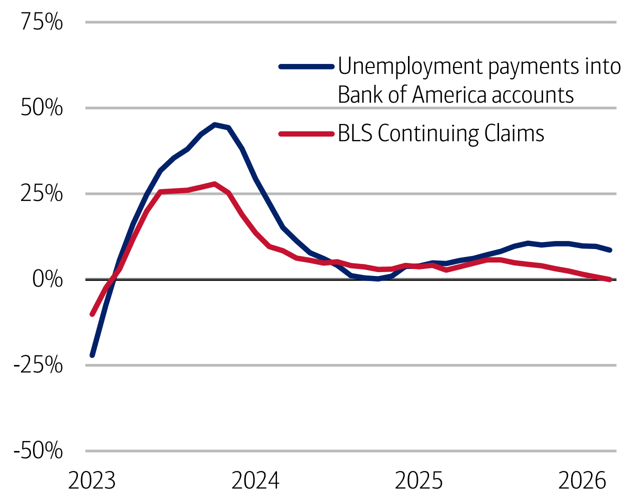
Exhibit 1: Bank of America account data indicates payrolls growth has rebounded to rates comparable to those observed in early 2025
Payroll estimates from Bank of America customer deposit account data (three-month moving average, % YoY), the Bureau of Labor Statistics (BLS) and Automatic Data Processing (ADP) (monthly, YoY)



Source: Bank of America internal data, Haver Analytics
Note: BLS and ADP data are seasonally adjusted, Bank of America data is not seasonally adjusted.

BANK OF AMERICA INSTITUTE

Exhibit 2: Unemployment payments growth into Bank of America customer accounts has slowed
Number of households receiving unemployment payments (three-month moving average, YoY%, not seasonally adjusted (NSA)) and Continuing Claims (three-month moving average, YoY%, seasonally adjusted (SA))



Source: Bank of America internal data, Bloomberg

BANK OF AMERICA INSTITUTE

The Bureau of Labor Statistics (BLS) payrolls growth figure shows considerably weaker YoY jobs growth than the estimate based on Bank of America data. In February, BLS payrolls dropped by 133K month-over-month (MoM), though the number of jobs was still up by 149K YoY. In the latest March data, the BLS estimates payrolls rose 178K MoM and are up 260K YoY, which appears directionally consistent with the stronger picture seen in Bank of America data.

Additionally, Bank of America data on unemployment payments into customer accounts shows growth has eased. In March, unemployment payments growth dropped below 9% YoY (Exhibit 2). This also appears directionally consistent with BLS Continuing Claims data, which shows YoY growth around zero. The overall household survey unemployment rate was 4.3% in March, down from 4.4% in February.

… but there is a very large gap in after-tax wage growth between higher-income households and other cohorts

While it appears jobs growth momentum has improved, there is a very large gap between after-tax wage and salary growth across income cohorts. In fact, Bank of America deposit data shows higher-income households' after-tax wage and salary growth rose to 5.6% YoY in March, up from 4.2% YoY in February. This is the highest after-tax wage growth for this cohort since August 2021 (Exhibit 3).

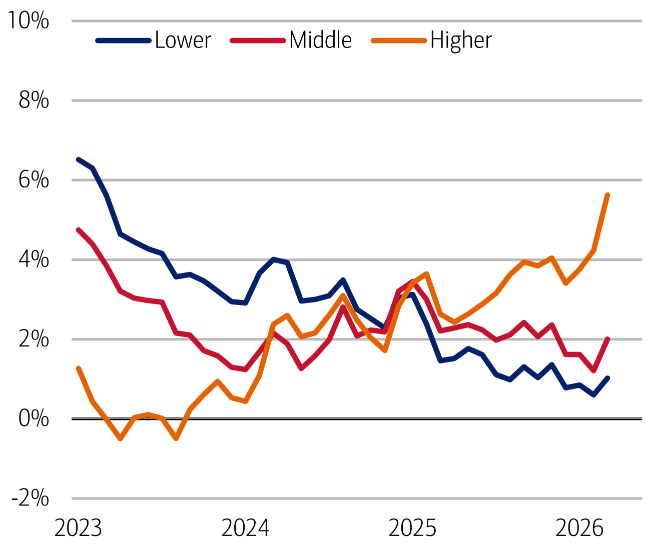
There was also a rebound in middle- and lower-income households' after-tax wage growth in March, to 2.0% and 1.0% YoY, respectively. But the gap between these cohorts' and higher-income households' wage growth still widened and is now the largest gap we've seen since the data series began in 2015.

In our view, one driver of the current widening in after-tax wage growth between higher-income households and the other cohorts is likely bonus growth. An estimate of bonus growth (see Methodology) based on Bank of America deposit data showed higher-income households saw solid positive growth over the first three months of 2026, while for middle- and lower-income households, bonus growth was negative (Exhibit 4).

This divergence in bonus growth may fade as we move through the year (many higher-income households receive their bonuses in the early part of the year). But, even so, the persistence in the "K" shape between higher-income wage growth and other cohorts suggests underlying wage dynamics may continue to favor the higher-income group.

Exhibit 3: In March, higher-income households' after-tax wage growth rose to 5.6% YoY, while lower-income households' wage growth was 1.0% YoY

After-tax wage and salary growth by household income terciles, based on Bank of America aggregated consumer deposit account data (3-month moving average, YoY%, SA)

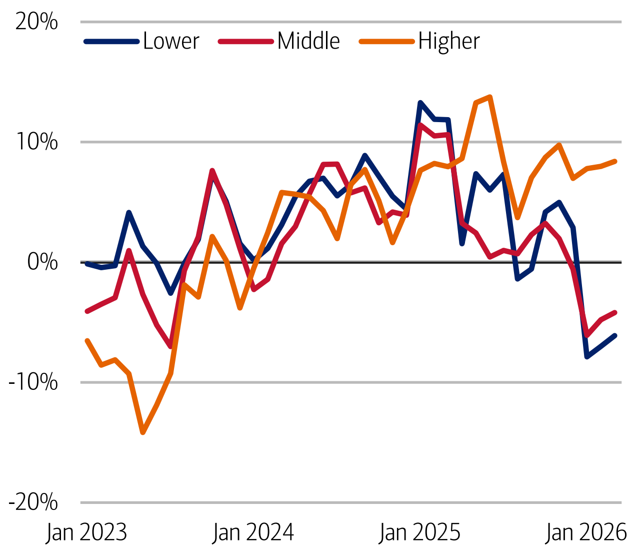


Source: Bank of America internal data

BANK OF AMERICA INSTITUTE

Exhibit 4: It appears there has been a large divergence in bonus growth in 2026

Estimate of average bonus payments by household income terciles (3-month moving average, YoY)



Source: Bank of America internal data

BANK OF AMERICA INSTITUTE

Methodology

Selected Bank of America transaction data is used to inform the macroeconomic views expressed in this report and should be considered in the context of other economic indicators and publicly available information. In certain instances, the data may provide directional and/or predictive value. The data used is not comprehensive; it is based on **aggregated and anonymized** selections of Bank of America data and may reflect a degree of selection bias and limitations on the data available.

Any payments data represents aggregated spend from US Retail, Preferred, Small Business and Wealth Management clients with a deposit account or credit card. Aggregated spend include total credit card, debit card, ACH, wires, bill pay, business/peer-to-peer, cash, and checks.

Any **Small Business** payments data represents aggregate spend from Small Business clients with a deposit account or a Small Business credit card. Payroll payments data include channels such as ACH (automated clearing house), bill pay, checks and wire. Bank of America per Small Business client data represents activity spending from active Small Business clients with a deposit account or a Small Business credit card and at least one transaction in each month. Small businesses in this report include business clients within Bank of America and generally defined as under $5mm in annual sales revenue.

Unless otherwise stated, data is not adjusted for seasonality, processing days or portfolio changes, and may be subject to periodic revisions.

The differences between the total and per household card spending growth rate (if discussed) can be explained by the following reasons:

1. Overall total card spending growth is partially boosted by the growth in the number of active cardholders in our sample. This could be due to an increasing customer base or inactive customers using their cards more frequently.

2. Per household card spending growth only looks at households that complete at least five transactions with Bank of America cards in the month. Per household spending growth isolates impacts from a changing sample size, which could be unrelated to underlying economic momentum, and potential spending volatility from less active users.

3. Overall total card spending includes small business card spending while per household card spending does not.

4. Differences due to using processing dates (total card spending) versus transaction date (per household card spending).

5. Other differences including household formations due to young adults moving in and out of their parent's houses during COVID.

Any household consumer deposit data based on Bank of America internal data is derived by anonymizing and aggregating data from Bank of America consumer deposit accounts in the US and analyzing that data at a highly aggregated level. Whenever median household savings and checking balances are quoted, the data is based on a fixed cohort of households that had a consumer deposit account (checking and/or savings account) for all months from January 2019 through the most current month of data shown.

Bank of America aggregated credit/debit card spending per household includes spending from active US households only. Only consumer card holders making a minimum of five transactions a month are included in the dataset. Spending from corporate cards are excluded. Data regarding merchants who receive payments are identified and classified by the Merchant Categorization Code (MCC) defined by financial services companies. The data are mapped using proprietary methods from the MCCs to the North American Industry Classification System (NAICS), which is also used by the Census Bureau, in order to classify spending data by subsector. Spending data may also be classified by other proprietary methods not using MCCs.

We consider a measure of services necessity spending that includes but is not limited to childcare, rent, insurance, public transportation, and tax payments. Discretionary services includes but is not limited to charitable donations, leisure travel, entertainment, and professional/consumer services. Holiday spending is defined as items in which spending in the November-December period is usually at least 20% of total annual spending on the category.

For analysis looking at higher value transactions (including durables), we consider a value per transaction threshold estimated with reference to the top 30% of transactions by value in 2024. The share of higher value transactions is then the number of transactions above this threshold as a percentage of total transactions over time.

Lower, middle and higher household income cuts in Bank of America credit and debit card spending per household, and consumer deposit account data are based on quantitative estimates of each households' income. These quantitative estimates are bucketed according to terciles, with a third of households placed in each tercile periodically. The lowest tercile represents 'lower income', the middle tercile represents 'middle income' and the highest tercile 'higher income'. The income thresholds between these terciles will move over time, reflecting any number of factors that impact income, including general wage inflation, changes in social security payments and individual households' income. The income and tercile in which a household is categorised are periodically re-assessed.

Generations, if discussed, are defined as follows:

1. Gen Z, born after 1995
2. Younger Millennials: born between 1989-1995
3. Older Millennials: born between 1978-1988
4. Gen Xers: born between 1965-1977
5. Baby Boomer: 1946-1964
6. Traditionalists: pre-1946

Any reference to card spending per household on gasoline includes all purchases at gasoline stations and might include purchases of non-gas items.

Estimate of payrolls growth from Bank of America internal data is based on the change in customer accounts receiving a paycheck in the month. An adjustment is made for the difference between overall population growth and customer account growth.

An estimate of bonus growth from Bank of America deposit data is calculated by looking at customers who have received an inbound ACH payroll transaction in the last two years. From this sample an estimate of bonuses is derived by looking for payroll transactions which are over 50% higher than the median regular payroll payments received by the customer. Of these payments only those that were received around the same time in each of the last two years are selected.

Additional information about the methodology used to aggregate the data is available upon request.

Contributors

David Michael Tinsley

Senior Economist, Bank of America Institute

Sources

Li Wei

Director, Global Risk Analytics

Yan Peng

Vice President, Global Risk Analytics

Akshita Jain

Assistant Vice President, Global Risk Analytics

Jasmine Elder

Vice President, Global Risk Analytics

Disclosures

These materials have been prepared by Bank of America Institute and are provided to you for general information purposes only. To the extent these materials reference Bank of America data, such materials are not intended to be reflective or indicative of, and should not be relied upon as, the results of operations, financial conditions or performance of Bank of America. Bank of America Institute is a think tank dedicated to uncovering powerful insights that move business and society forward. Drawing on data and resources from across the bank and the world, the Institute delivers important, original perspectives on the economy, sustainability and global transformation. Unless otherwise specifically stated, any views or opinions expressed herein are solely those of Bank of America Institute and any individual authors listed, and are not the product of the BofA Global Research department or any other department of Bank of America Corporation or its affiliates and/or subsidiaries (collectively Bank of America). The views in these materials may differ from the views and opinions expressed by the BofA Global Research department or other departments or divisions of Bank of America. Information has been obtained from sources believed to be reliable, but Bank of America does not warrant its completeness or accuracy. These materials do not make any claim regarding the sustainability of any product or service. Any discussion of sustainability is limited as set out herein. Views and estimates constitute our judgment as of the date of these materials and are subject to change without notice. The views expressed herein should not be construed as individual investment advice for any particular person and are not intended as recommendations of particular securities, financial instruments, strategies or banking services for a particular person. This material does not constitute an offer or an invitation by or on behalf of Bank of America to any person to buy or sell any security or financial instrument or engage in any banking service. Nothing in these materials constitutes investment, legal, accounting or tax advice.

Economy

Golf spending is slightly above par

02 April 2026

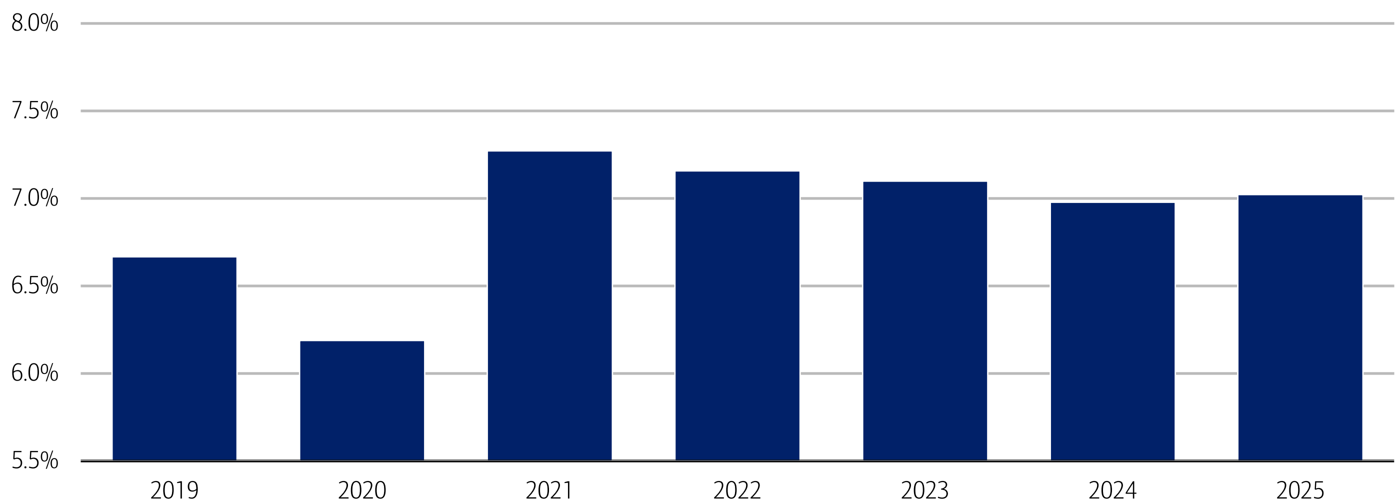
Key takeaways

- After three years of declining golf participation, from the 2021 pandemic peak, Bank of America payments data shows a modest rebound in 2025. At the same time, average annual spend per golfer has increased for four consecutive years, suggesting a smaller but more enthusiastic group of participants.
- Gen Z has continued to gain share in golf participation, likely driven by lower-cost, more accessible formats like driving ranges and simulators. At the same time, participation among younger Millennials has declined, potentially reflecting competing life priorities and tighter budgets - though those who remain active appear to be spending more.
- Regionally, the West has recorded the strongest golf spending growth by region in 2025, supported by domestic migration, destination-style "stay-to-play" courses, and expanded offerings. Other regions are seeing new players enter the sport, but typically through lower-cost golfing options.

Golf teed up a comeback in 2025

Golf began climbing out of the rough in 2025, reversing a three-year decline in the proportion of households paying for golf activities (e.g., golf courses, driving ranges, mini golf and simulators but not country club dues), according to Bank of America aggregated credit and debit card data (Exhibit 1). And our data suggests that, despite the pullback from the pandemic peak, participation remains above 2019 levels.

Exhibit 1: The share of households with golf spending peaked in 2021 and declined until 2024 before stabilizing in 2025
Share of US households with golf spending (yearly, %)



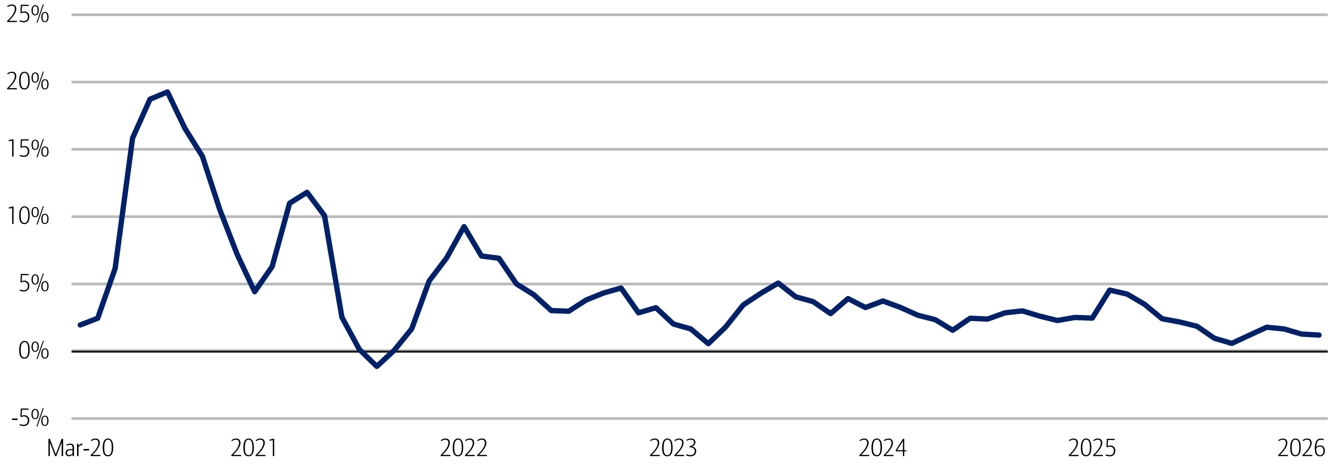
Source: Bank of America internal data. Note: Golf spending includes spending at golf courses, mini golf, driving ranges, and simulators but does not include country club dues or golf retailers.
BANK OF AMERICA INSTITUTE

Furthermore, most people who play golf seem to be deepening their relationship with the sport. Bank of America internal data found that per household spending continued to grow over the past four years, although it slowed to around 1% year-over-year (YoY) in February 2026 (Exhibit 2).

According to the National Golf Foundation, the number of rounds played has increased for the past three years, despite a decrease in the actual number of golf courses nationwide. Golf has also potentially received a boost from the ease and accessibility of driving ranges and indoor simulators. So overall, while the share of US households that play golf has held fairly steady over the past two years, the households that do play seem to be playing a bit more often.

Exhibit 2: Of the households that spend on golf, outlays have increased through February 2026, although at a slightly slower rate

Golf spending per household (3-month moving average, YoY%)



Source: Bank of America internal data

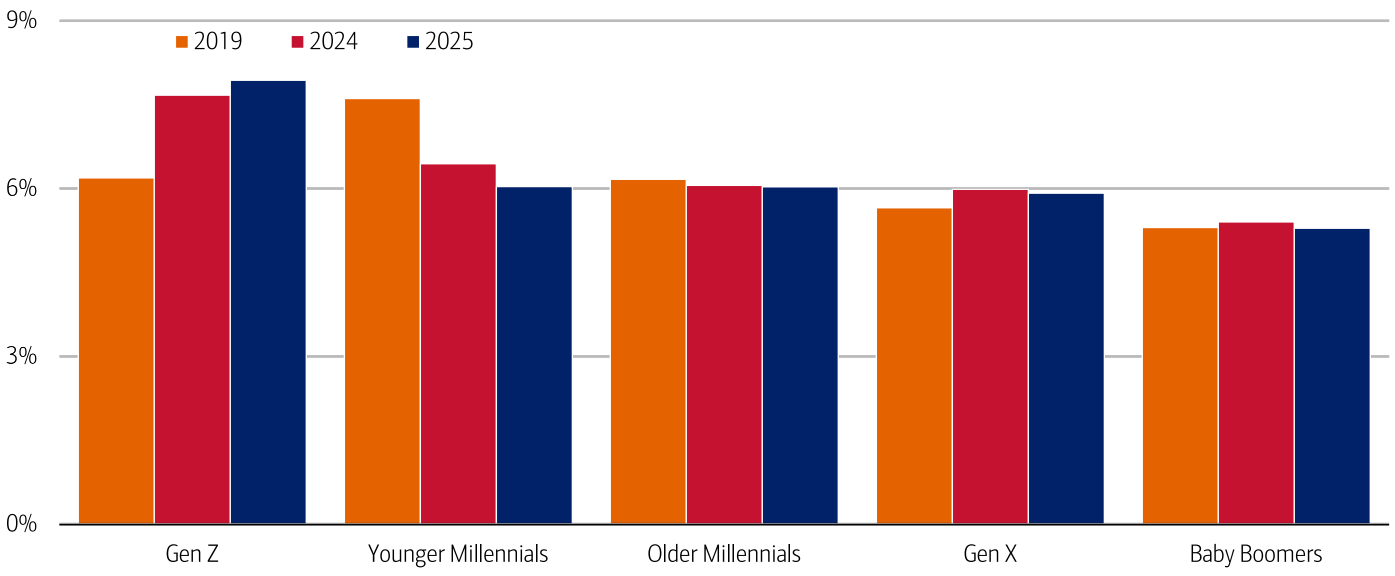
BANK OF AMERICA INSTITUTE

Passing the club: Millennials ease, as Gen Z drives adoption

Looking at golf spending by age, there's been a substantial drop in the share of younger Millennials participating in the sport, along with a smaller decrease among older Millennials since 2019 (Exhibit 3). Some in this generation may be pulling back as they take on more life responsibilities, such as starting families or building their careers. Other may be pulling back on discretionary spending as they took on more financial responsibilities during a period of elevated costs – driven by inflation and higher-than-usual interest rates (read more in [Autos: stuck in a lower gear?]).

Exhibit 3: There's been a significant increase in the share of Gen Z households spending on golf since the pandemic

Share of households with golf spending by age generation (yearly, %)



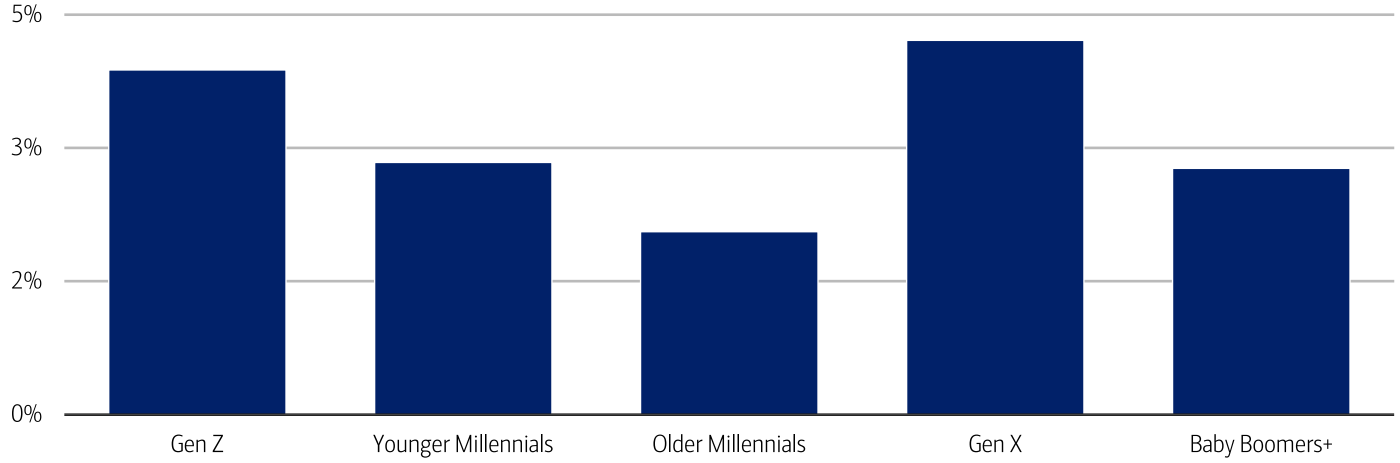
Source: Bank of America internal data

BANK OF AMERICA INSTITUTE

Conversely, there's been a significant increase in the share of Gen Z households spending on golf. In our view, this could reflect the somewhat recent popularity of driving ranges and indoor simulators, as well as their desire to participate in more healthy experiences (read more in [Younger generations move from barstools to barbells]). It could also be that Gen Z came of age during a period of heightened social distancing, which may have encouraged adoption of outdoor sports like golf.

Furthermore, looking just at households that spent on golf in 2025, golf spending growth was faster among Gen Z but still slower than Gen X (Exhibit 4), likely reflecting the latter's stronger overall financial position, allowing more spend and more time on the course.

Exhibit 4: Gen Z are catching up with Gen X in terms of fastest spending growth
Spending growth on golf by generation (2025 compared to 2024, YoY%)



Source: Bank of America internal data

BANK OF AMERICA INSTITUTE

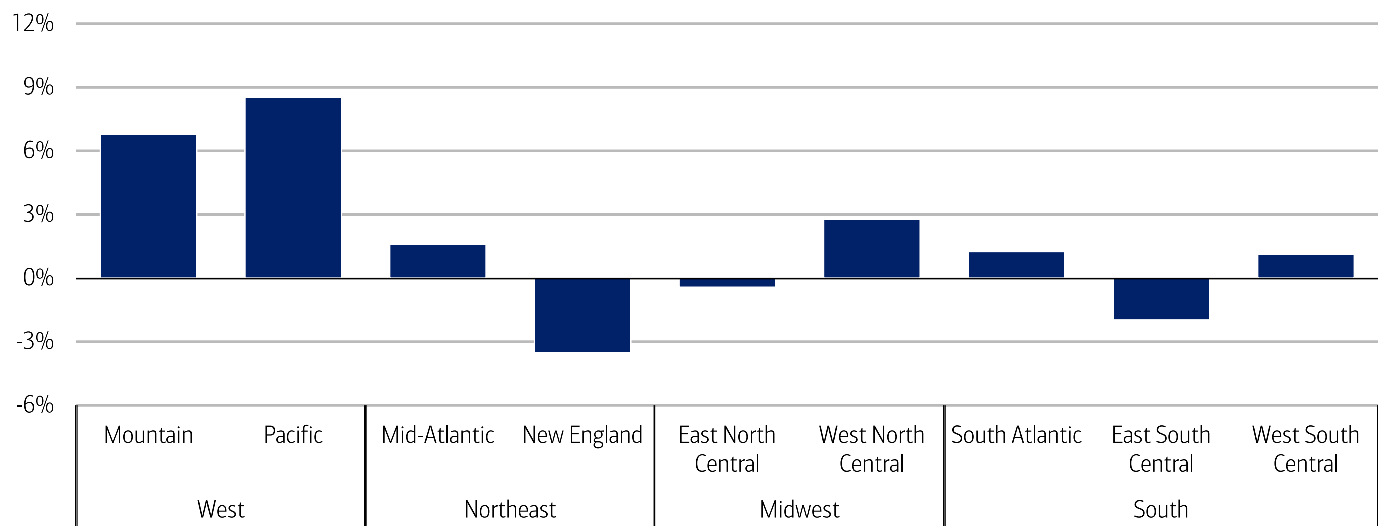
Golf heads West

Looking across the US, the West is clearly above par for golf. Spending increased around 7% and 9% YoY per customer in the Mountain (Arizona, Colorado, Utah – see Methodology for full list of states) and Pacific (California, Washington, etc.) divisions, respectively (Exhibit 5). The picture was mixed in all other regions.

In our view, more spending in the West may reflect some of the increases in domestic migration to the Western portion of the Sunbelt (read more in On the move: US migration patterns). Another part of the story may be the rise in "stay-to-play" policies, where golfers stay at the course and pay course fees. Also, some golf resorts in the region have added new courses that might incentivize some golfers to expand their length of stay and spend more money.

Other divisions including the West North Central (e.g., Minnesota, Nebraska, Missouri, etc.) and the West South Central (Texas, Louisiana, etc.) have experienced an increase in the number of residents playing golf, although new players are gravitating toward less expensive golfing experiences.

Exhibit 5: Golf spending in the West rose at least twice as fast YoY than any other region in 2025
Aggregate spending by US Census region and division (2025 compared to 2024, YoY%)



Source: Bank of America internal data

BANK OF AMERICA INSTITUTE

Methodology

Selected Bank of America transaction data is used to inform the macroeconomic views expressed in this report and should be considered in the context of other economic indicators and publicly available information. In certain instances, the data may provide directional and/or predictive value. The data used is not comprehensive; it is based on **aggregated and anonymized** selections of Bank of America data and may reflect a degree of selection bias and limitations on the data available.

Any payments data represents aggregated spend from US Retail, Preferred, Small Business and Wealth Management clients with a deposit account or credit card. Aggregated spend include total credit card, debit card, ACH, wires, bill pay, business/peer-to-peer, cash, and checks.

Any **Small Business** payments data represents aggregate spend from Small Business clients with a deposit account or a Small Business credit card. Payroll payments data include channels such as ACH (automated clearing house), bill pay, checks and wire. Bank of America per Small Business client data represents activity spending from active Small Business clients with a deposit account or a Small Business credit card and at least one transaction in each month. Small businesses in this report include business clients within Bank of America and generally defined as under $5mm in annual sales revenue.

Unless otherwise stated, data is not adjusted for seasonality, processing days or portfolio changes, and may be subject to periodic revisions.

The differences between the total and per household card spending growth rate (if discussed) can be explained by the following reasons:

1. Overall total card spending growth is partially boosted by the growth in the number of active cardholders in our sample. This could be due to an increasing customer base or inactive customers using their cards more frequently.

2. Per household card spending growth only looks at households that complete at least five transactions with Bank of America cards in the month. Per household spending growth isolates impacts from a changing sample size, which could be unrelated to underlying economic momentum, and potential spending volatility from less active users.

3. Overall total card spending includes small business card spending while per household card spending does not.

4. Differences due to using processing dates (total card spending) versus transaction date (per household card spending).

5. Other differences including household formations due to young adults moving in and out of their parent's houses during COVID.

Any household consumer deposit data based on Bank of America internal data is derived by anonymizing and aggregating data from Bank of America consumer deposit accounts in the US and analyzing that data at a highly aggregated level. Whenever median household savings and checking balances are quoted, the data is based on a fixed cohort of households that had a consumer deposit account (checking and/or savings account) for all months from January 2019 through the most current month of data shown.

Bank of America aggregated credit/debit card spending per household includes spending from active US households only. Only consumer card holders making a minimum of five transactions a month are included in the dataset. Spending from corporate cards are excluded. Data regarding merchants who receive payments are identified and classified by the Merchant Categorization Code (MCC) defined by financial services companies. The data are mapped using proprietary methods from the MCCs to the North American Industry Classification System (NAICS), which is also used by the Census Bureau, in order to classify spending data by subsector. Spending data may also be classified by other proprietary methods not using MCCs.

We consider a measure of services necessity spending that includes but is not limited to childcare, rent, insurance, insurance, public transportation, and tax payments. Discretionary services includes but is not limited to charitable donations, leisure travel, entertainment, and professional/consumer services. Holiday spending is defined as items in which spending in the November-December period is usually at least 20% of total annual spending on the category.

Golf spending includes spending at courses, mini golf, driving ranges, and simulators but does not include country club dues or golf retailers.

Lower, middle and higher household income cuts in Bank of America credit and debit card spending per household, and consumer deposit account data are based on quantitative estimates of each households' income. These quantitative estimates are bucketed according to terciles, with a third of households placed in each tercile periodically. The lowest tercile represents 'lower income', the middle tercile represents 'middle income' and the highest tercile 'higher income'. The income thresholds between these terciles will move over time, reflecting any number of factors that impact income, including general wage inflation, changes in social security payments and individual households' income. The income and tercile in which a household is categorised are periodically re-assessed.

US Census Regions of the United States:

Northeast: New Jersey, Pennsylvania, New York (Mid-Atlantic); Maine, Connecticut, Massachusetts, New Hampshire, Rhode Island, Vermont (New England)

Midwest: Indiana, Wisconsin, Illinois, , Michigan, Ohio (East North Central); Missouri, Minnesota, Iowa, Nebraska, North Dakota, South Dakota, Kansas

South: Delaware, Texas, Louisiana, Arkansas, Oklahoma (West South Central); Mississippi, Alabama, Tennessee, Kentucky (East South Central); Washington DC, Florida, Georgia, Maryland, North Carolina, South Carolina, Virginia, West Virginia (South Atlantic)

West: Arizona, Colorado, Idaho, Montana, Nevada, New Mexico, Utah, Wyoming (Mountain); Alaska, California, Hawaii, Oregon, Washington (Pacific)

Major grocery categories include sugar and sweets, juices and other non-alcoholic beverages, bakery products, processed fruits and vegetables, fresh fruit and vegetables, coffee and tea, fats and oils, milk, cereal and cereal products, other, cheese, and meats, poultry and fish, Other includes soups, snacks, frozen and freeze-dried prepared foods, and spices, seasonings, and condiments.

Generations, if discussed, are defined as follows:

1. Gen Z, born after 1995
2. Younger Millennials: born between 1989-1995
3. Older Millennials: born between 1978-1988
4. Gen Xers: born between 1965-1977
5. Baby Boomer: 1946-1964
6. Traditionalists: pre-1946

Any reference to card spending per household on gasoline includes all purchases at gasoline stations and might include purchases of non-gas items.

Additional information about the methodology used to aggregate the data is available upon request.

Contributors

Joe Wadford

Economist, Bank of America Institute

Sources

Jon Kaplan

Senior Vice President, Consumer & Small Business Products and Analytics

Disclosures

Sustainability

Secondhand fashion creates a closet refresh

21 April 2026

Key takeaways

- After nearly three years of declines, clothing spending per household has grown steadily since August 2025, rising 5.1% year-over-year (YoY) in March on a three-month moving average, according to Bank of America credit and debit card data. However, most of the growth is concentrated in luxury fashion and discount apparel, while department stores continue to lose ground, signaling increased evidence of a "K"-shaped trend in apparel.
- The number of secondhand fashion transactions per household grew nine times faster than secondhand spending in March, yet consumers across all income groups are spending less on each purchase since April 2025, according to Bank of America credit and debit card data. In our view, this points to consumers turning to resale more often to stretch budgets and shop sustainably.
- The number of Bank of America customers selling secondhand clothing grew 16% YoY in March, with Gen Z accounting for 41% of sellers year-to-date in 2026. While the fastest growth is among those selling more than four times a month, the majority still transact infrequently, suggesting resale is more often a closet clean-out than a true income stream for now.

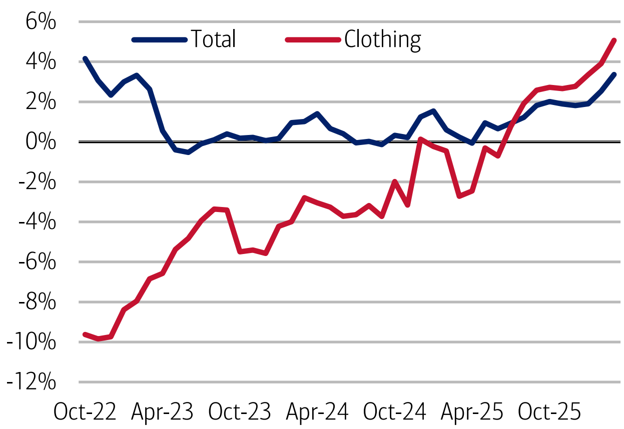
Buying clothes is back in fashion

After almost three years of declines, spending growth per household on clothing as measured by Bank of America aggregated credit and debit card data has shown a largely steady trajectory of positive year-over-year gains since August 2025 (Exhibit 1). More recently, overall clothing spending rose 5.1% year-over-year (YoY) in March on a non-seasonally adjusted (NSA) three-month moving average, surpassing total card spending.

Not all clothing purchases are in style: per Bank of America card data, luxury fashion saw the biggest gains in Q1 2026, more than five time higher than growth in Q4 2025 (Exhibit 2). Conversely, teen retail apparel (targeted for 13-19-year-olds) and department stores declined further from Q4 2025.

Exhibit 1: Spending growth for clothing has trended above total card spending growth since August 2025

Bank of America credit and debit card spending per household by category (non-seasonally adjusted, YoY%, monthly, 3-month moving average)

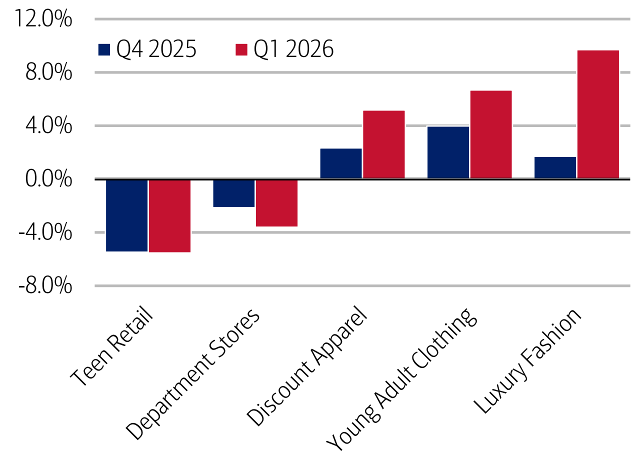


Source: Bank of America internal data

BANK OF AMERICA INSTITUTE

Exhibit 2: Department store spending growth fell 3.6% YoY in Q1 2026, worsening from the previous quarter

Credit and debit card spending per household by apparel category (NSA, quarterly average, YoY%)



Source: Bank of America internal data

BANK OF AMERICA INSTITUTE

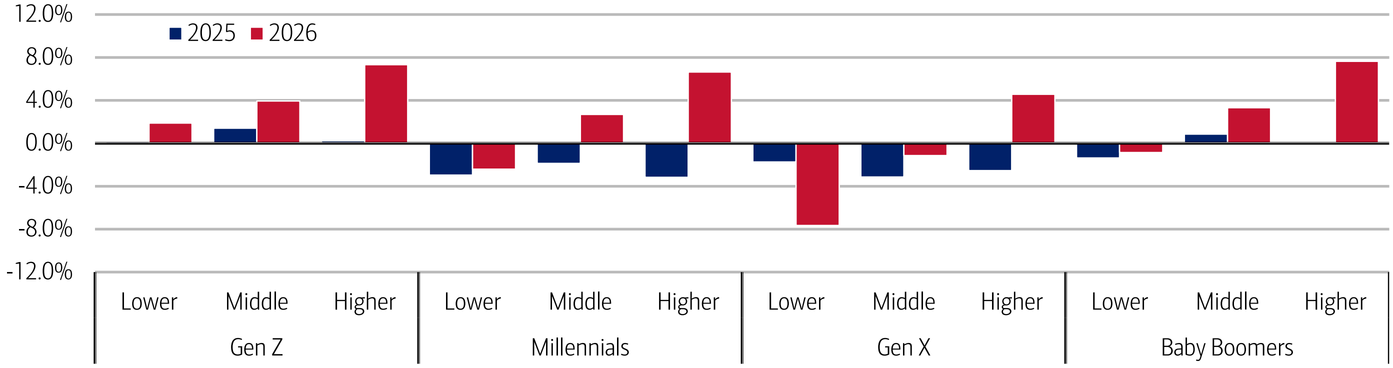
Gen Z fit check evident even among the "K"

Discount apparel spending growth also increased along with the category of young adult (targeted for 20–24-year-olds) clothing. The discount segment is likely benefitting in part from tax refunds as well as trading down, while the gains in young adult clothing could be supported by Gen Z, whose YoY apparel spending growth overall was up across all income cohorts – the only generation to do so (Exhibit 3).

The strength among higher-income consumers provides further proof of the "K"-shaped economy, especially when it comes to discretionary spending (read more on this in our [April Consumer Checkpoint]). By contrast, the only group whose spending growth deteriorated this year compared to last was lower-income Gen X households (read more on generational differences in spending in [March's Will younger-gen spending hit a gas-price speed bump?]).

Exhibit 3: Gen Z is the only generation to have increased apparel spending growth in March across all income cohorts

Clothing spending per household by income and generation in March (annual, YoY%)



Source: Bank of America internal data

BANK OF AMERICA INSTITUTE

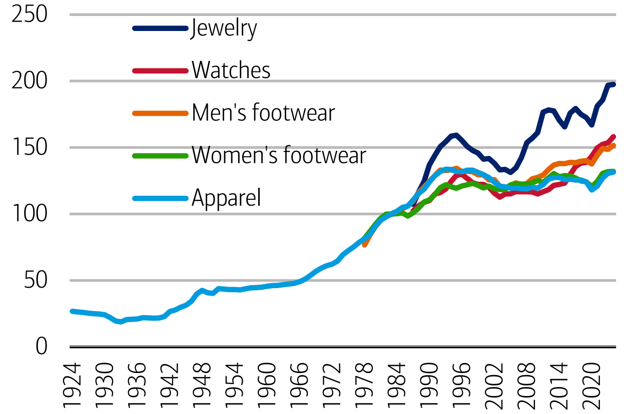
Secondhand fashion offers an environmentally friendly and economic alternative

Still, as inflation persists, and retailers confront tariff costs, consumers are facing higher price tags on apparel – about five times higher than they were a century ago (Exhibit 4). Beyond discount apparel, secondhand fashion offers consumers an environmentally friendly and economical alternative to clothing purchases. Using Bank of America credit and debit card spending data, we found transaction growth per household on secondhand fashion (see Methodology for definition) was nine times that of secondhand spending growth in March (Exhibit 5), suggesting consumers believe that dollars go further when shopping that way.

Note that, for this secondhand fashion analysis, the scope of this series is limited to market-driven secondhand fashion platforms and does not extend to donation-based, charity-centered thrift models. Additionally, spending is predominantly online but can include brick-and-mortar.

Exhibit 4: Apparel prices are about 5x higher than a century ago

Consumer price index for apparel categories (annual, index, 1982-84 = 100)

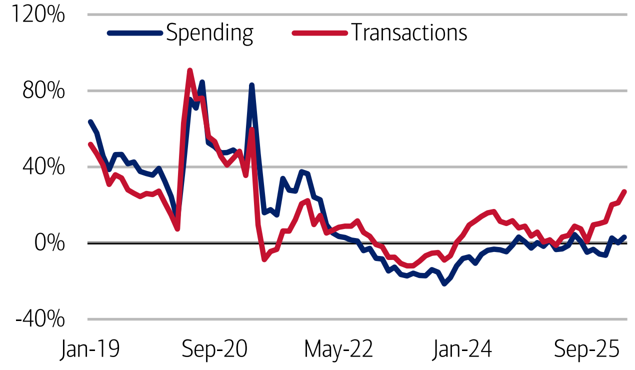


Source: Bureau of Labor Statistics

BANK OF AMERICA INSTITUTE

Exhibit 5: Secondhand fashion transactions growth per household was up 22% YoY in March

Secondhand fashion spending and transaction growth per household (monthly, YoY%)



Source: Bank of America internal data

Note: Secondhand fashion is comprised of market-driven secondhand fashion platforms and does not extend to donation-based, charity-centered thrift models.

BANK OF AMERICA INSTITUTE

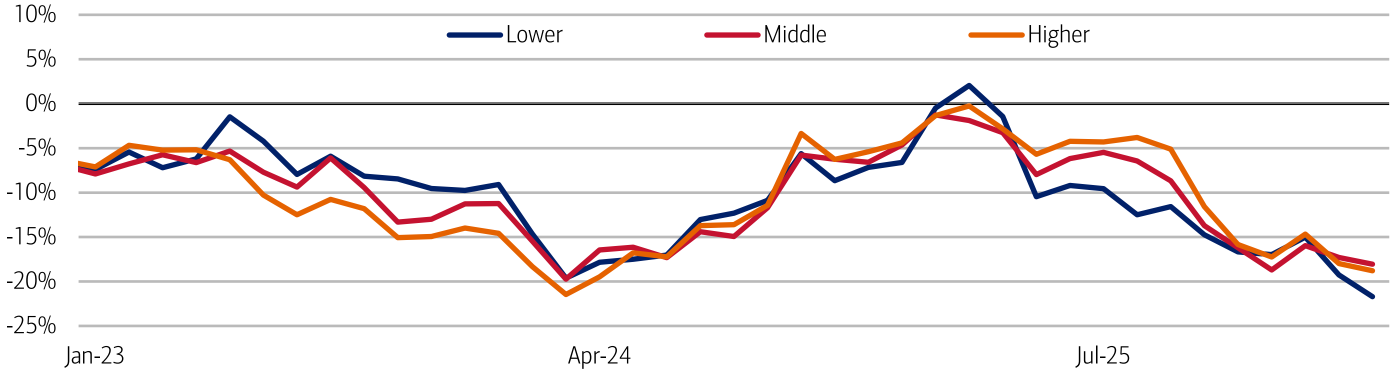
BANK OF AMERICA INSTITUTE

Higher-income Millennials spend more on secondhand fashion

Higher-income households account for the greatest share of dollars spent in March on secondhand clothing, according to Bank of America debit and credit card data. And by generation, Millennials spent the most, followed by Gen X and Gen Z.

Additionally, spending per transaction growth for secondhand fashion has been falling for the past six months across all income cohorts (Exhibit 6), indicating that shoppers are spending less on each purchase, especially lower-income households.

Exhibit 6: Spending per transaction on secondhand fashion has been falling further across all income groups since April 2025
Secondhand fashion spending per transaction growth (monthly, YoY%)



Source: Bank of America internal data
Note: Secondhand fashion is comprised of market-driven secondhand fashion platforms and does not extend to donation-based, charity-centered thrift models.

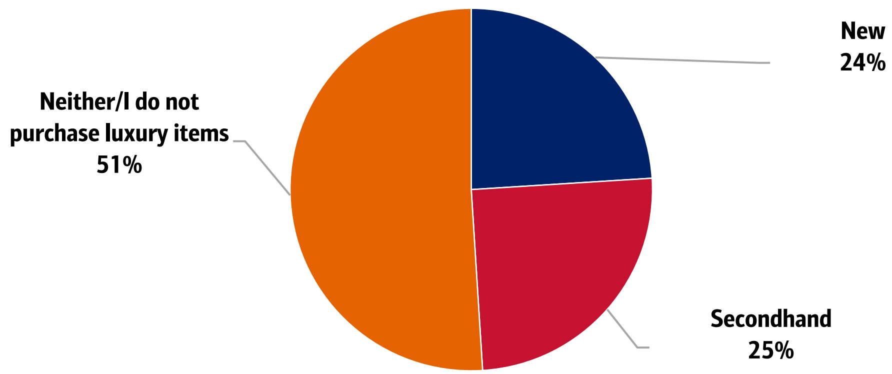
BANK OF AMERICA INSTITUTE

"Recommerce," sustainable luxury fashion, and a circular economy

Rising demand for "circular" goods – previously used items - is driving spending growth in resale marketplaces like secondhand clothing. The US secondhand apparel market is expected to reach $78.8 billion by 2030, growing at an average annual rate of about 7.3%, according to online resale marketplace ThredUp's 2026 Resale Report conducted by GlobalData.[1] And, in 2025, the US secondhand apparel market grew nearly four times faster than the broader retail clothing market, per the report, driven by growing consumer preferences for value, wardrobe variety and sustainability.

According to BofA Global Research, on average, pre-loved clothes sell for between 30% and 40% of their original retail price. Furthermore, for consumers seeking luxury goods, there is a more active secondary market for many items, according to BofA Global Research. This reflects changing consumer tastes and a growing acceptance of pre-owned luxury (Exhibit 7).

Exhibit 7: Secondhand markets open doors to luxury brands, with as many respondents say they purchase secondhand items as those who do new
Do you prefer to purchase luxury apparel and accessories new or second-hand? (% of respondents)



Source: CivicScience
Note: 5,855 Responses from March 22, 2024 to July 11, 2025

BANK OF AMERICA INSTITUTE

[1] ThredUp and GlobalData. (2026, April 2). *2026 Resale Report.* ThredUp.

Gen Z turns secondhand into a side hustle

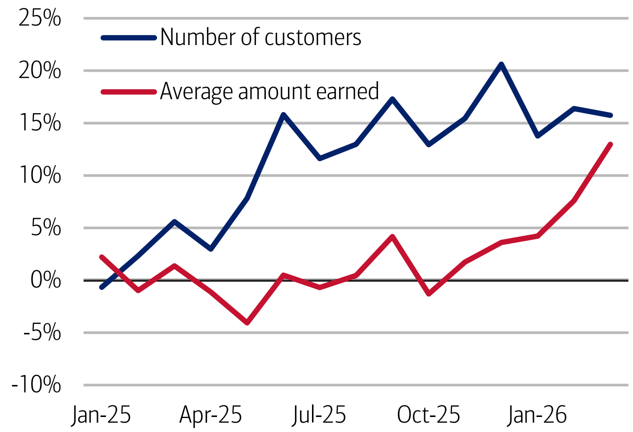
Consumers aren't just in secondhand markets to spend money – they also want to earn it. Using Bank of America internal data, we identify payment inflows from secondhand retailers into customer accounts through debit card and/or Automated Clearing House (ACH) channels to analyze who is earning extra income from secondhand shopping. Note this data excludes revenue yet to be transferred to a customer's account (i.e. keeping money from a sale on a platform as credit towards another purchase).

In March, the number of Bank of America customers selling at secondhand retailers grew 16% YoY (Exhibit 8). This trend has mostly been on the rise since the start of 2025. Interestingly, the average monthly earnings (i.e. revenue per customer), was up 13% YoY in March – a notable increase from the previous six months, where growth was largely flat.

Who are the sellers? Gen Z comprised 41% of secondhand sellers year-to-date (YTD) in 2026, up from 37% in 2024 (Exhibit 9). Millennials averaged 38%, followed by Gen X at 14%.

Exhibit 8: The number of customers selling at secondhand retailers increased 16% YoY in March

Secondhand sellers and average amount earned (monthly, YoY%)



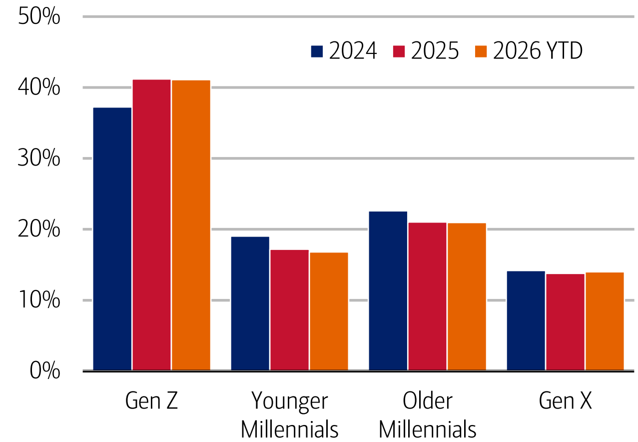
Source: Bank of America internal data

Note: Secondhand fashion is comprised of market-driven secondhand fashion platforms and does not extend to donation-based, charity-centered thrift models.

BANK OF AMERICA INSTITUTE

Exhibit 9: Gen Z comprised 41% of secondhand sellers year-to-date in 2026

Secondhand sellers by generation (% of customers, annual average)



Source: Bank of America internal data

Note: Secondhand fashion is comprised of market-driven secondhand fashion platforms and does not extend to donation-based, charity-centered thrift models.

BANK OF AMERICA INSTITUTE

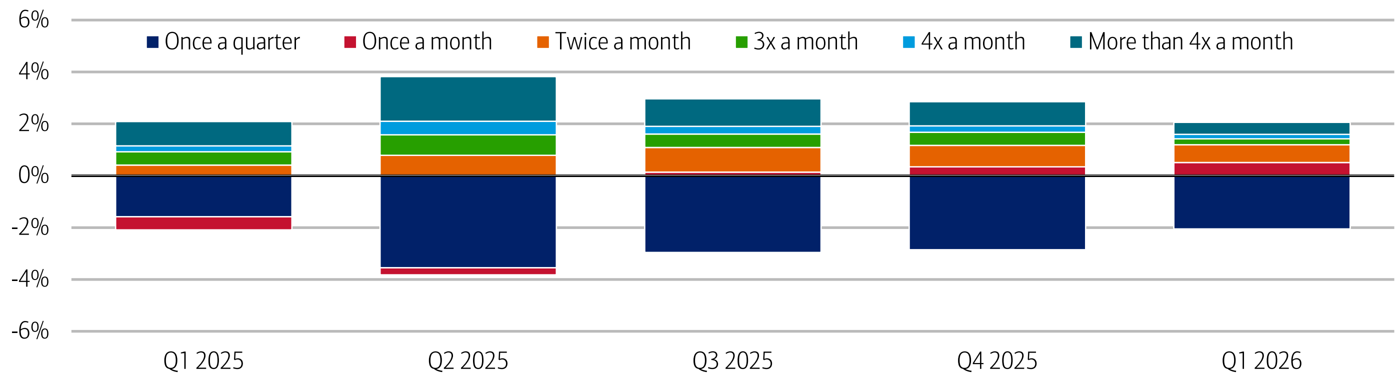
Almost three-quarters of sellers only do so once a quarter or once a month

In Bank of America internal data, most customers selling on secondhand marketplaces offer items once a quarter or once a month, with almost 73% of them following this pattern in Q1 2026. This suggests to us that selling on these platforms is still predominantly used as a way to make some extra pocket change or participate in sustainability.

However, secondhand sellers are increasingly transacting more frequently, with average growth in share over the past four quarters greatest amongst those customers, selling more than four times a month (Exhibit 10).

Exhibit 10: Sellers are increasingly transacting more than once a quarter

Frequency in which customers sell secondhand (percentage point difference from last year's period)



Source: Bank of America internal data

Note: Secondhand fashion is comprised of market-driven secondhand fashion platforms and does not extend to donation-based, charity-centered thrift models.

BANK OF AMERICA INSTITUTE

Methodology

Selected Bank of America transaction data is used to inform the macroeconomic views expressed in this report and should be considered in the context of other economic indicators and publicly available information. In certain instances, the data may provide directional and/or predictive value. The data used is not comprehensive; it is based on **aggregated and anonymized** selections of Bank of America data and may reflect a degree of selection bias and limitations on the data available.

Bank of America credit/debit card spending per household includes spending from active US households only. Only consumer card holders making a minimum of five transactions a month are included in the dataset. Spending from corporate cards is excluded. Data regarding merchants who receive payments are identified and classified by the Merchant Categorization Code (MCC) defined by financial services companies. The data are mapped using proprietary methods from the MCCs to the North American Industry Classification System (NAICS), which is also used by the Census Bureau, in order to classify spending data by subsector. Spending data may also be classified by other proprietary methods not using MCCs.

If applicable, the consumer deposit data based on Bank of America internal data is derived by anonymizing and aggregating data from Bank of America consumer deposit accounts in the US and analyzing that data at a highly aggregated level.

If applicable, any payments data represents aggregated spend from US Retail, Preferred, Small Business and Wealth Management clients with a deposit account or credit card. Any reference to aggregated AI spend include total credit card, debit card, ACH, or bill pay.

Secondhand fashion transactions are conducted through Bank of America debit card or ACH channels and secondhand retailers are identified using industry-researched merchant names. Note that the scope of this series is limited to market-driven secondhand fashion platforms and does not extend to donation-based, charity-centered thrift models. A majority of these transactions were conducted on digital platforms but could also include brick and mortar transactions.

Generations, if discussed, are defined as follows:

1. Gen Z, born after 1995

2. Younger Millennials: born between 1989-1995

3. Older Millennials: born between 1978-1988

4. Gen Xers: born between 1965-1977

5. Baby Boomer: 1946-1964

6. Traditionalists: pre-1946

CivicScience polls were conducted to various audiences across the US. These are unweighted responses and are conducted through the specific time frame given.

Unless otherwise stated, data is not adjusted for seasonality, processing days or portfolio changes, and may be subject to periodic revisions.

Additional information about the methodology used to aggregate the data is available upon request.

Contributors

Taylor Bowley

Economist, Bank of America Institute

Sources

Jasmine Elder

Vice President, Global Risk Analytics

Jon Kaplan

Senior Vice President, Analytics, Modeling and Insights

William M. Reuter

Research Analyst, BofA Global Research

Disclosures

These materials have been prepared by Bank of America Institute and are provided to you for general information purposes only. To the extent these materials reference Bank of America data, such materials are not intended to be reflective or indicative of, and should not be relied upon as, the results of operations, financial conditions or performance of Bank of America. Bank of America Institute is a think tank dedicated to uncovering powerful insights that move business and society forward. Drawing on data and resources from across the bank and the world, the Institute delivers important, original perspectives on the economy, sustainability and global transformation. Unless otherwise specifically stated, any views or opinions expressed herein are solely those of Bank of America Institute and any individual authors listed, and are not the product of the BofA Global Research department or any other department of Bank of America Corporation or its affiliates and/or subsidiaries (collectively Bank of America). The views in these materials may differ from the views and opinions expressed by the BofA Global Research department or other departments or divisions of Bank of America. Information has been obtained from sources believed to be reliable, but Bank of America does not warrant its completeness or accuracy. These materials do not make any claim regarding the sustainability of any product or service. Any discussion of sustainability is limited as set out herein. Views and estimates constitute our judgment as of the date of these materials and are subject to change without notice. The views expressed herein should not be construed as individual investment advice for any particular person and are not intended as recommendations of particular securities, financial instruments, strategies or banking services for a particular person. This material does not constitute an offer or an invitation by or on behalf of Bank of America to any person to buy or sell any security or financial instrument or engage in any banking service. Nothing in these materials constitutes investment, legal, accounting or tax advice.

Accessible version

BANK OF AMERICA INSTITUTE

BANK OF AMERICA

Transformation

Feeding the world with AI

07 April 2026

Key takeaways

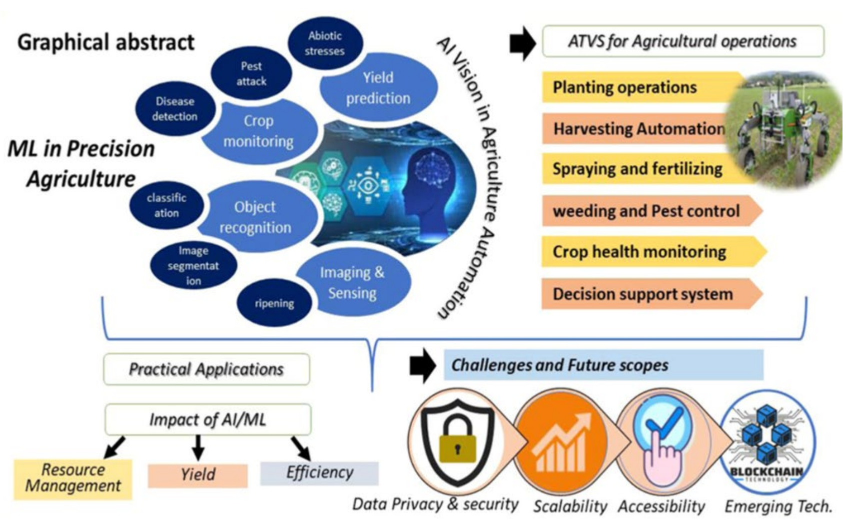
- Agriculture is undergoing its biggest technological shift in decades as AI becomes embedded across soil management, irrigation, fertilization and crop monitoring. By 2024, over half of farmers had adopted or were willing to adopt AI-enabled tools, driven by measurable gains in decision-making, yields, efficiency and sustainability.
- Digital AI has transformed how farmers understand their fields, but insight alone is no longer sufficient amid climate volatility, labor shortages, rising input costs and non-linear yield risks - including geopolitical instability along critical fertilizer supply corridors. These pressures demand not just better decisions, but precise, timely, plant-level action - something traditional advisory AI cannot deliver.
- That execution gap is pulling the sector toward physical AI, which enables real-time, plant-by-plant control. This shift from "AI for advice" to "AI for autonomous agronomy" directly moves revenue, cost and risk curves in a sector defined by tight margins and biological variability.

Agriculture's latest technological evolution

Agriculture, a sector that represents roughly 4% of global gross domestic product (GDP), is undergoing its biggest technological shift in decades, according to BofA Global Research. AI has already become deeply embedded in modern farming practices, improving farmers' decisions around soil, irrigation, fertilization, crop monitoring and disease detection (Exhibit 1). And the use of sensors, drones, satellite imaging and predictive crop models creates a rich layer of digital intelligence that can help farmers understand their fields with unprecedented clarity.

Exhibit 1: Crop monitoring + resource management = higher productivity for sustainable agriculture

Infographic illustrating machine learning (ML) in precision agriculture



Source: www.sciencedirect.com CC-by 4.0[1]; BofA Global Research

Note: ATVs = all-terrain vehicles

BANK OF AMERICA INSTITUTE

[1] Kumar, A., Kumar, R., Padhiary, M., Saha, D., & Sethi, L.N. (2024, June 3). *Enhancing precision agriculture: A comprehensive review of machine learning and AI vision applications in all-terrain vehicle for farm automation.* ScienceDirect. https://doi.org/10.1016/j.atech.2024.100483

AI moves from early adoption to sector-wide penetration

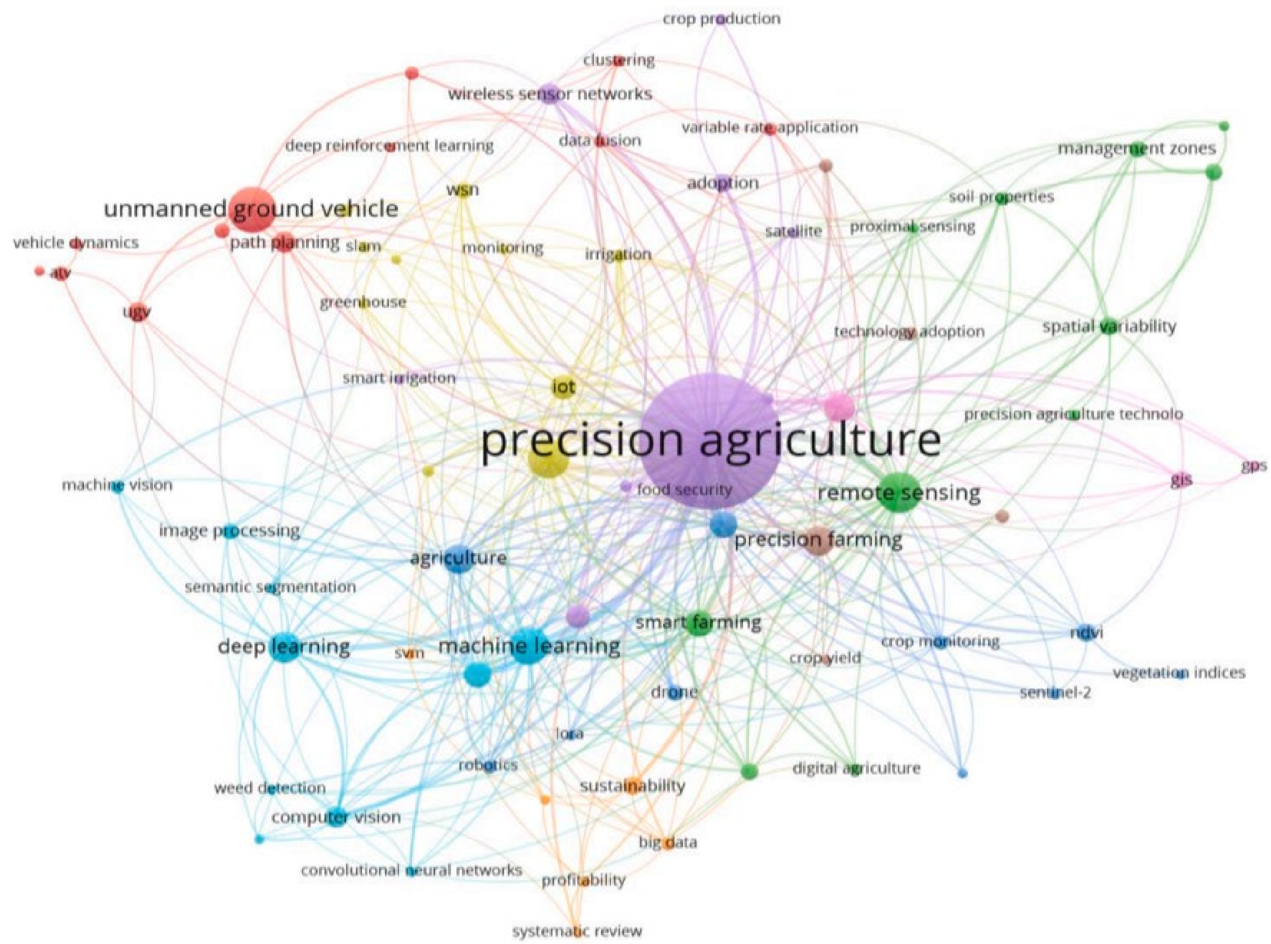
As of 2024, over half of farmers worldwide had adopted or were willing to adopt at least one precision-agriculture or AI-enabled technology.[2] This uptake is driven by tangible agronomic benefits (related to crop growing and soil management): 80% of farmers using data analytics report better decision-making. AI-enabled precision irrigation and fertilization can raise crop yields by 25%.[3] Additionally, AI improves water usage efficiency and fertilizer application accuracy,[4] while also reducing emissions[5] – an increasingly important advantage when fertilizer supplies are constrained – thereby promoting sustainable farming practices.

AI-in-agriculture market: ~$47 billion by 2034E

Applications of AI in agriculture now span an increasingly diverse set of technologies, from AI-guided vertical farming and biological enhancement tools, to AI-based crop modeling and precision agriculture using drones, sensors and satellites for real-time field intelligence. Money flows reflect this shift: agricultural technology (agtech) companies raised $7 billion in 2025, up nearly 4% year-over-year (YoY), with precision-agriculture companies outpacing deals for crop inputs and enhancements.[6]

Exhibit 2: Deploying precision agriculture for farm automation is attracting growing interest from researchers and industry stakeholders

Keyword network diagram for precision agriculture



Source: www.sciencedirect.com CC-by 4.0[7], BofA Global Research

Note: The keyword network diagram based on 77 most occurring keywords out of 2998, with number of occurrences of a keyword being ≥ 5 (Based on 1055 publication accessed from Scopus database with keywords 'ML in agriculture, AI vision, ATV, Precision Agriculture and unmanned ground vehicle')(2024, March 17)

BANK OF AMERICA INSTITUTE

[2] Statista Research Department. (2024, October). *Share of farmers using or willing to adopt at least one new technology in the US in 2024, by technology type.* Statista.

[3] Ferdusmou, J., Hossain, M.A, Khatoon, R., & Saha, S. (2025, January). *Smart Farming Revolution: AI-Powered Solutions for Sustainable Growth and Profit*. Journal of Management World.

[4] Ibid.

[5] Hoque, A., Padhiary, M., Roy, S., & Saikia, P. (2025, March 18). *Climate-Smart Agriculture: AI-Based Solutions for Enhancing Crop Resilience and Reducing Environmental Impact*. Asian Research Journal of Agriculture 18(1): 291-310. https://doi.org/10.9734/arja/2025/v18i1665

[6] Frederick, A. (2026, January 6). *Agrifood VC First Look.* PitchBook.

[7] Kumar, A., Kumar, R., Padhiary, M., Saha, D., & Sethi, L.N. (2024, June 3). *Enhancing precision agriculture: A comprehensive review of machine learning and AI vision applications in all-terrain vehicle for farm automation.* ScienceDirect. https://doi.org/10.1016/j.atech.2024.100483

Broader market outlooks underscore this transition. The AI-in-agriculture market is forecasted to increase at a 26.3% compound annual growth rate (CAGR) to $46.6 billion by 2034, per Global Market Insights.[8] This is driven by increased use of precision inputs, labor substitution and real-time agronomic decision support. Machine learning – now representing roughly half of the market – underpins emerging technologies such as generative AI, autonomous tractors and robotic sprayers (read AI dictionary, part 1: The basics for an intro to machine learning). It also enables real-time object recognition across key agricultural applications including weed detection, livestock monitoring and yield estimation from aerial imagery.

According to BofA Global Research, North America currently leads agricultural AI adoption with a 36% market share, supported by strong capex appetite and a mature precision-agriculture ecosystem that speeds up deployment of autonomous equipment. All these trends show that farming is shifting from *digital* agronomy (decision support) to *autonomous* agronomy (execution) (Exhibit 2). Furthermore, this expansion of AI-driven and robotics-based farming technology is supported by favorable regulation, which simplifies deployment. Falling hardware costs for sensors, compute and machine vision also make adoption more affordable. At the same time, policy incentives are increasingly aligned with technologies that verifiably reduce chemical inputs, accelerating uptake.

Physical AI: Brains behind the wheel

Physical AI integrates artificial intelligence into physical systems, such as humanoid robots, autonomous vehicles and drones (we further define this topic in Physical AI, part 1: The basics). As a result, AI can perceive, reason and act in the real world – not just generate digital outputs. This matters because the next productivity wave won't come from better dashboards, but from execution at the edge, delivered by robots, drones, autonomous implements and sensor-driven machines that can continuously close the loop between sensing and action, per BofA Global Research.

Capital and capability are accelerating

Over 70 companies are now developing physical AI infrastructure across data, simulation, foundation models and observability. Robotics and physical AI companies raised ~$41 billion in 2025, while equity deals into robot foundation model developers alone increased more than tenfold from three in 2021 to 32 in 2025. Collectively, these advances position physical AI as a shared, reusable infrastructure, deployed across robots, vehicles and drones, rather than an isolated, task-specific solution.

The shift is structural: Climate volatility is tightening constraints of farming

According to BofA Global Research, agriculture is a sector where physical AI can scale in the near term because the underlying operating environment has changed. Fertilizer markets are tightly locked into a handful of regions and energy routes, so even localized geopolitical disruptions can quickly ripple through global input costs.

And on the climate front, drought and heat stress are no longer episodic; they are persistent conditions that directly cap yields and destabilize broader farm economics (Exhibit 3). The European Union's Joint Research Centre found that 52 individual prolonged meteorological drought events occurred from August 2023 to July 2024.[9] Droughts, heatwaves and warm spells have negatively impacted crop productivity across Europe, Southern Africa, Central and Southern America and Southeast Asia. In this regime, many crops exhibit non-linear responses to temperature. Yields can improve gradually up to a threshold, then deteriorate sharply once exceeded. Meanwhile, rising heat accelerates evapotranspiration (loss of water to the atmosphere from soil evaporation and plant transpiration), pulling crops into water stress faster and worsening pest and disease pressure.

Why "advisory AI" hits a ceiling

Digital AI has transformed how farmers understand their fields, but insight alone is no longer enough. The pressures reshaping global agriculture are climate volatility, chronic labor shortages, rising input costs and non-linear yield risks. These require not just better decisions, but precise, timely, plant-level action. This is where traditional "advisory" AI reaches its ceiling. A farmer can know a disease outbreak is emerging, yet without the labor, equipment or precision to intervene quickly and locally, the insight can't be monetized. Modern agriculture now needs systems that can see, decide and act in the real world. That execution gap is what is pulling the sector toward physical AI.

Physical AI addresses the gap by enabling real-time, plant-by-plant control – dynamic irrigation, targeted nutrient delivery, early pest intervention and precision application based on live perception. This marks a shift from "AI for advice" to "AI for autonomous agronomy." In a sector defined by narrow margins and biological variability, that shift directly moves revenue, cost and risk curves.

Economics is driving commercial scale

Physical AI is scaling because the economics have turned. Technologies that deliver payback in under a year scale fastest in agriculture and falling hardware costs are strengthening this equation. Precision robotics that reduce labor, chemical use and operational time can pay back in months rather than years. Yield stability in a more volatile climate becomes an additional economic engine. The shift is from automation as a way to reduce costs to a means to boost profitability and resilience.

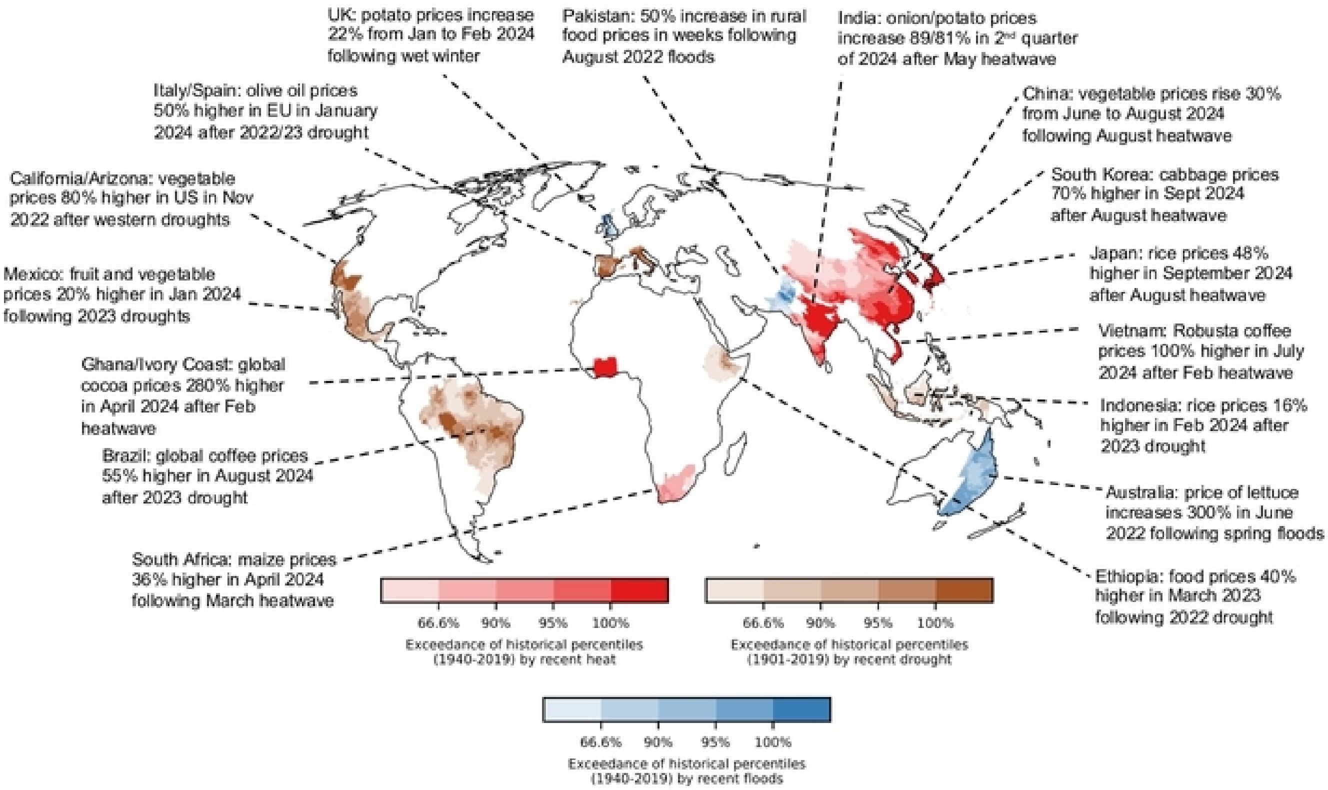
---

[8] Global Market Insights. (2025, May). *AI in Agriculture Market Size & Share 2025-2034.*

[9] Joint Research Centre. (2024, October 2). *Global drought threatens food supplies and energy production*. European Commission.

Exhibit 3: Food prices have spiked across the globe in response to extreme climate conditions

Map of climate-induced food price spikes since 2022



Source: ResearchGate CC-by 4.0[10], BofA Global Research

Note: Underlying shading represents the extent to which the associated climate conditions exceeded percentiles of the historical distribution.

BANK OF AMERICA INSTITUTE

[10] Donat, M.G., Kotz, M., Lancaster, T., Parker, M., Smith, P., Taylor, A., & Vetter, S. (2025, July). *Climate extremes, food price spikes, and their wider societal risks*. Environmental Research Letters. https://doi.org/10.1088/1748-9326/ade45f

Contributors

Lynelle Huskey

Analyst, Bank of America Institute

Vanessa Cook

Content Strategist, Bank of America Institute

Sources

Menka Bajaj

Sustainability Strategist, BofA Global Research

Haim Israel

Equity Strategist, BofA Global Research

Disclosures